Exact Affine Geometry, Optimal Centres, and Certified Compression Bounds for Finite-Outcome Quantum Instruments

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Abstract

Quantum instruments describe the full statistics of a measurement together with the conditional post-measurement state, and they form the building blocks of adaptive protocols and quantum networks. We study the geometry of the set of all n-outcome instruments acting between two finite-dimensional quantum systems, and we ask how many continuous real parameters must be retained in order to encode an arbitrary instrument and later reconstruct it with uniformly small error, under the convention of continuous encoding and unrestricted decoding. The geometry of the instrument body is determined exactly: its affine dimension is $d_A^2(nd_B^2 - 1)$; the largest diamond-norm ball contained in the body has radius $2/(nd_B \min\{d_A, d_B\})$ and is centred at the uniform depolarising instrument, which is optimal among all centres; and the same instrument is optimal for the covering problem, with an exact covering radius linked to the in-radius by a complementarity identity. Every radius and width carries a single-shot operational meaning through the optimal discrimination error. On the compression side, antipodal constructions — output preparation, an input-dependent observable sphere, and a full-direction section — produce a certified lower envelope on the optimal error; constructive upper bounds from constant codes, convex projection, and balanced pinching meet the envelope exactly at zero latent dimension (where both equal the exact constant-code width), in the collapse case, and from the affine dimension on, and bracket it in the intermediate regime; the zero-error threshold equals the affine dimension. The single-outcome case recovers quantum channels, and the one-dimensional-output case recovers finite-outcome measurements; the trade-off between the number of outcomes and the robustness of the uniform instrument is made explicit.

Keywords: quantum instruments; diamond norm; compression width; antipodal profile; Chebyshev radius; quantum measurements; no-programming theorem.

1 Introduction

Quantum instruments are the natural generalisation of quantum channels to finite-outcome statistics: a collection of completely positive, trace-nonincreasing maps whose sum is a channel. They describe both the probability of an outcome and the conditional post-measurement state, and they model finite-outcome measurements, adaptive protocols, and the nodes of quantum networks [5, 7, 9, 10].

The general source–latent theory underlying the obstruction arguments — fiber radii, antipodal profiles, deleted-product coindex, and stable Lipschitz variants — is developed for channels in Ref. [2]; the present paper uses the specialisations recorded in Sections 4 and 5. Throughout, the $n = 1$ restrictions of the instrument theorems below recover the corresponding channel-level statements of [2]: the observable sphere, the full-direction sphere, the join, the global in-radius, and the covering results.

The question treated here is the instrument-level analogue of the channel problem: does the compact convex body of n-outcome instruments between A and B admit a continuous low-dimensional latent encoding with uniformly accurate decoding? Throughout, we adopt the standard asymmetric convention: the *encoder is continuous* and the *decoder is unrestricted* (Convention 5.1); where a continuous decoder is additionally available we say so explicitly. The answer has three regimes. At small latent dimension, input-independent preparation of a flagged output state already forces error one. In an intermediate range, the width is bracketed between a certified lower and a certified upper envelope: a centred-ball bound scaling as $1/n$ — which Theorem 3.12 shows is the best centred-ball certificate at any centre — plus a full-direction antipodal obstruction contributing an n-independent component $1/d_A$, against upper bounds that never sink below one. From the affine dimension on, exact reconstruction holds, even with a continuous decoder.

The main contribution of this paper is the exact affine geometry of the instrument body and its compression consequences: the affine dimension, the centred and global in-radii, the covering radius, and the diameter are determined exactly, with the uniform depolarising instrument turning out to be an optimal centre simultaneously for the in-radius and the covering radius (a concave–convex duality, Theorem 3.12 and Remark 3.13). Which quantities are determined *exactly*, and which only by *certified bounds*, is stated once in Remark 2.6 and maintained throughout. A separate physical question — whether an abstract code can be realised by a fixed programmable processor — is addressed in Appendix A: exact finite-dimensional evaluation of all unitary channels is impossible, and vanishing error forces unbounded program dimension. This separates *representational* dimension from *programmable* evaluation; the main theorems do not rely on the appendix. For the input-independent case $d_A = 1$, the flat-width problem is governed by the flag-coordinate count nd_B: the antipodal floor 1 is exact precisely on the upper side $nd_B \leq 2r + 2$ of the subcritical range for classical and qubit outputs, and the width is bounded away from 1 on the lower side $nd_B \geq 2r + 3$ — at least $4/3$, with the exact boundary value $4/3$ and the exact value $2(1 - 1/d_B)$ at latent dimension $n - 1$ for prime-power output dimensions; and the bottom width is classified completely — flat exactly on the classical $n \leq 4$ and qubit $n \leq 2$ bodies, at least $4/3$ on every other input-independent body, including every state space of dimension ≥ 3, with the qutrit state space exactly at $4/3$ (Theorems 6.1, 6.2, 6.8, 6.10, 6.12, 6.13, 6.14; Corollaries 6.7 and 6.16).

The paper is organised as follows. Section 2 fixes notation and conventions; the specialised forms of the source–latent theory of Ref. [2] are recorded where they are used, in Sections 4 and 5. Section 3 determines the exact affine and metric parameters of the instrument body, and Section 4 constructs the antipodal certificates and the certified lower envelope. Section 5 develops the constructive upper bounds and the exact thresholds, and Section 6 collects the specialisations, worked examples, and the outcome trade-off. Section 7 summarises which quantities are exact and which are certified and lists the open problems; Appendix A records the no-programming background.

2 Preliminaries

Throughout, A and B are finite-dimensional Hilbert spaces of dimensions d_A, d_B, and $n \in \mathbb{N}$ is the number of outcomes. Let $A_0 \simeq A$ be a reference copy with a fixed pair of conjugate bases in A_0 and A, and define the maximally entangled state

$$|\omega_A\rangle = \frac{1}{\sqrt{d_A}} \sum_{j=1}^{d_A} |j\rangle_{A_0} |j\rangle_A.$$

For a map $\Phi : \mathrm{L}(A) \to \mathrm{L}(B)$ the *normalised Choi operator* is

$$\widehat{J}_\Phi = (\mathrm{id}_{A_0} \otimes \Phi)(|\omega_A\rangle\langle\omega_A|),$$

the Choi–Jamiołkowski isomorphism in the normalised form [1, 3].

Remark 2.1 (Normalised Choi convention). *With the unnormalised maximally entangled vector* $|\Omega_A\rangle = \sum_j |j\rangle_{A_0}|j\rangle_A$ *one has* $\widehat{J}_\Phi = \frac{1}{d_A} J_\Phi^{\mathrm{unnorm}}$. *Consequently* Φ *is trace preserving if and only if*

$$\operatorname{Tr}_B \widehat{J}_\Phi = \frac{I_{A_0}}{d_A}.$$

All affine constraints, radii, covering bounds, and pinching dimension counts in this paper use this normalised convention.

Let C_n be an n-dimensional classical flag space with basis $|1\rangle, \dots, |n\rangle$.

Definition 2.2 (Instrument, flagged channel, and instrument diamond norm). *An* n*-outcome instrument from* A *to* B *is a tuple* $\mathcal{I} = (\mathcal{I}_1, \dots, \mathcal{I}_n)$ *of completely positive, trace-nonincreasing maps whose sum is a channel. Write* $\mathrm{Inst}_n(A, B)$ *for the set of all such instruments. The flagged channel of* $\mathcal{I}$ *is*

$$\widehat{\mathcal{I}}(X) = \sum_{k=1}^{n} \mathcal{I}_k(X) \otimes |k\rangle\langle k|.$$

For a tuple $\vec{\Delta} = (\Delta_1, \dots, \Delta_n)$ *of Hermiticity-preserving maps, define the flagged map and the instrument diamond norm by*

$$\widehat{\vec{\Delta}}(X) = \sum_{k=1}^{n} \Delta_k(X) \otimes |k\rangle\langle k|, \qquad \|\vec{\Delta}\|_{\diamond,\mathrm{inst}} = \|\widehat{\vec{\Delta}}\|_\diamond.$$

By block diagonality (the trace norm of a block-diagonal operator is the sum of the trace norms of its blocks) this equals the operational formula

$$\|\vec{\Delta}\|_{\diamond,\mathrm{inst}} = \sup_{\rho \in \mathcal{S}(A_0 \otimes A)} \sum_{k=1}^{n} \|(\mathrm{id}_{A_0} \otimes \Delta_k)(\rho)\|_1.$$

The diamond norm was introduced in [6]; a reference system of input dimension d_A *suffices for it [12]. The summands are the unconditional, generally subnormalised outcome contributions. The body* $\mathrm{Inst}_n(A, B)$ *is compact and convex: convexity follows from the convexity of the completely positive cone and the affine normalisation constraints, and compactness from positivity together with the summed marginal condition, which bounds every Choi block.*

Remark 2.3 (Operational meaning). *For two instruments* $\mathcal{I}, \mathcal{J}$ *with equal prior probabilities, the optimal single-use discrimination success probability, with arbitrary reference assistance, is*

$$p_{\mathrm{succ}} = \frac{1}{2} + \frac{1}{4} \|\mathcal{I} - \mathcal{J}\|_{\diamond,\mathrm{inst}},$$

i.e. the advantage over random guessing is $\frac{1}{4}\|\mathcal{I} - \mathcal{J}\|_{\diamond,\mathrm{inst}}$ *(an instance of the Holevo–Helstrom theorem for binary state discrimination [11], applied to the flagged outputs). Thus every radius and width stated in the instrument diamond norm has a direct single-shot discrimination interpretation, up to this conventional factor of* $1/4$.

The ambient affine space and its direction space are

$$\mathfrak{A}_{A,B}^{(n)} = \Big\{(\Phi_1, \dots, \Phi_n) : \Phi_k \text{ Hermiticity-preserving}, \quad \sum_{k=1}^{n} \operatorname{Tr}_B \widehat{J}_{\Phi_k} = \frac{I_{A_0}}{d_A}\Big\},$$

$$\mathfrak{V}_{A,B}^{(n)} = \Big\{(\Delta_1, \dots, \Delta_n) : \Delta_k \text{ Hermiticity-preserving}, \quad \sum_{k=1}^{n} \operatorname{Tr}_B \widehat{J}_{\Delta_k} = 0\Big\}.$$

Here $\mathfrak{V}^{(n)}_{A,B}$ is the real linear direction space, of real dimension $D^{(n)}_{A,B}$ when $nd_B > 1$. Thus only the summed direction is trace-annihilating; individual outcome components need not be. For $\mathcal{I}_0 \in \mathfrak{A}^{(n)}_{A,B}$ define the closed relative instrument-diamond ball

$$\overline{B}^{\mathrm{rel}}_{\diamond,\mathrm{inst}}(\mathcal{I}_0, \rho) = \big\{\mathcal{I}_0 + \vec{\Delta} : \vec{\Delta} \in \mathfrak{V}^{(n)}_{A,B},\ \|\vec{\Delta}\|_{\diamond,\mathrm{inst}} \le \rho\big\},$$

and, for $\mathcal{I}_0 \in \mathrm{Inst}_n(A,B)$, its centred relative in-radius

$$\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) = \sup\Big\{\rho \ge 0 : \overline{B}^{\mathrm{rel}}_{\diamond,\mathrm{inst}}(\mathcal{I}_0, \rho) \subseteq \mathrm{Inst}_n(A,B)\Big\}.$$

Let $\mathcal{I}_*$ be the uniform depolarising instrument, $(\mathcal{I}_*)_k = \Omega_{A,B}/n$, where $\Omega_{A,B}(X) = \mathrm{Tr}(X) I_B/d_B$, so that $\widehat{J}_{(\mathcal{I}_*)_k} = I_{A_0 \otimes B}/(nd_A d_B)$. The centred in-radius of the body about $\mathcal{I}_*$ and its covering radius, denoted ρ_n and R_n throughout with $s = \min\{d_A, d_B\}$, are determined exactly in Theorems 3.4 and 3.8 below: $\rho_n = 2/(nd_B s)$ and $R_n = 2 - \rho_n$.

Remark 2.4 (POVM case $d_B = 1$). *When $d_B = 1$, identify $\mathrm{L}(B)$ with $\mathbb{C}$. An instrument component is a positive functional $\mathcal{I}_k(X) = \mathrm{Tr}(E_k X)$ for a unique effect $E_k \ge 0$, and the instrument condition is exactly $\sum_{k=1}^n E_k = I_A$. Thus $\mathrm{Inst}_n(A,\mathbb{C})$ is the convex body of n-outcome POVMs on A.*

Remark 2.5 (Degenerate singleton case). *If $n = d_B = 1$, then $\mathrm{Inst}_1(A,\mathbb{C})$ consists of the unique trace functional $X \mapsto \mathrm{Tr}(X)$: its affine dimension is $D^{(1)}_{A,1} = 0$, its direction space is $\{0\}$, every compression width is 0, and, because every relative ball is the singleton, the centred relative in-radius is $+\infty$ under the supremum convention (the formula $2/(nd_B s)$ does not apply). Except where explicitly stated, all nontrivial statements in this paper assume*

$$nd_B > 1.$$

Remark 2.6 (Exact claims versus certified bounds). *The following statements are exact: (i) the affine dimension $D^{(n)}_{A,B}$; (ii) the centred in-radius at the uniform instrument; (iii) the global in-radius over all centres, which equals (ii) by Theorem 3.12; (iv) the global covering radius and optimality of $\mathcal{I}_*$ as a centre; (v) the diameter, equal to 2; (vi) the antipodal profile of the replacement body, and hence of the full body when $d_A = 1$; (vii) the constant-code width $\delta^{\diamond}_{0,\mathrm{inst}}(A,B) = R_{\mathrm{inst}}(\mathcal{I}_*)$, which supplies the $r = 0$ value of the certified lower envelope (Definition 4.17); (viii) the zero-error Euclidean threshold for continuous encoding with unrestricted decoding; (ix) the full width function in the collapse case $nd_B s = 2$ (Theorem 5.18); (x) for $d_A = d_B = 1$, the full subcritical dichotomy: $\delta_r = 1$ for $n \le 2r+2$, $\delta_r \ge 4/3$ for $n \ge 2r+3$, with the exact boundary value $\delta_r = 4/3$ at $n = 2r+3$ (Theorem 6.8, Corollaries 6.9 and 6.7); (xi) for $d_A = 1$, $d_B = 2$, the full subcritical dichotomy $\delta_r = 1$ for $n-1 \le r \le nd_B^2 - 2$ and $\delta_r \ge 4/3$ for $r \le n-2$ (Theorem 6.10, Corollaries 6.4 and 6.11); (xii) for $d_A = 1$ and d_B a prime power, the exact width $\delta_{n-1} = 2(1 - 1/d_B)$, and the exact bottom width $\delta_1 = 4/3$ at $n = 5$ (Theorem 6.12, Corollary 6.7); (xiii) for $d_A = 1$, the complete bottom-width classification: $\delta_1 = 1$ exactly when $nd_B \le 4$ and $d_B \le 2$, $\delta_1 \ge 4/3$ otherwise, with the exact value $4/3$ at $(n, d_B) \in \{(5,1),(1,3),(2,3)\}$, the one-outcome ququart in $[4/3, 3/2]$, and $\delta_1 = 4/3$ on every state space of dimension ≥ 3 (Theorems 6.13, 6.14 and 6.12; Corollaries 6.7 and 6.16). The following are certified bounds or open quantities, not exact determinations: (a) the intermediate compression widths $\delta^{\diamond}_{r,\mathrm{inst}}(A,B)$ for $0 < r < D^{(n)}_{A,B}$ outside the input-independent flat regimes of items (x)–(xiii); (b) the full antipodal profile beyond the certified constructions; (c) optimality of the certified upper envelope $U^{(n)}_r$. Table 1 collects the complete regime-by-regime picture.*

Remark 2.7 (Notation). *To avoid collisions, the following symbols are used as consistently as the subject allows: ranks are denoted r_E, the number of input pinching blocks by k_{in}, the number of output pinching blocks by ℓ_{out}, the latent dimension by r (or q for general latent spaces), the* *covering radius by R, subspaces of input systems, flag-output systems, or supports by W (the join constructions of Section 4 use $W \subseteq B \otimes C_n$ on the flag-output side), and the reference system in the covering arguments by E (a copy of A_0, the diamond-norm reference of the definition of the instrument diamond norm); Appendix A additionally uses E for the programming input system, in a self-contained notation. The conjugations in Lemma 3.2 and Theorem 3.4 act on opposite factors of the reference pair; both are consistent with the fixed Choi bases and produce the same maximally entangled probes.*

3 Exact affine and metric parameters

This section determines the affine dimension, the centred in-radius at the uniform instrument, the global in-radius, the covering radius, and the diameter of the instrument body, all exactly.

Proposition 3.1 (Affine dimension of the instrument body). *Assume $nd_B > 1$. Then*

$$\dim_{\mathbb{R}} \operatorname{aff} \operatorname{Inst}_n(A, B) = D^{(n)}_{A,B} := d_A^2(nd_B^2 - 1).$$

Proof. An instrument Choi tuple consists of n Hermitian operators on $A_0 \otimes B$, of total real dimension $nd_A^2 d_B^2$. The normalisation constraint

$$\sum_{k=1}^{n} \operatorname{Tr}_B \widehat{J}_{\mathcal{I}_k} = \frac{I_{A_0}}{d_A}$$

imposes $d_A^2 = \dim_{\mathbb{R}} \operatorname{Herm}(A_0)$ real affine equations. The linear map $(X_1, \ldots, X_n) \mapsto \sum_k \operatorname{Tr}_B X_k$ from $\bigoplus_k \operatorname{Herm}(A_0 \otimes B)$ to $\operatorname{Herm}(A_0)$ is surjective: for $H \in \operatorname{Herm}(A_0)$ set $X_1 = H \otimes I_B/d_B$ (so $\operatorname{Tr}_B X_1 = H$) and $X_{k\geq 2} = 0$. Hence the constraints are independent. The tuple of $\mathcal{I}_*$ has every block $\widehat{J}_{(\mathcal{I}_*)_k} = I_{A_0 \otimes B}/(nd_A d_B)$ *positive definite*; every sufficiently small Hermitian perturbation in $\mathfrak{V}^{(n)}_{A,B}$ therefore remains positive semidefinite. Hence $\mathcal{I}_*$ is a relative interior point, positivity introduces no further affine equalities, and the affine count $nd_A^2 d_B^2 - d_A^2 = d_A^2(nd_B^2 - 1)$ is attained. □

Lemma 3.2 (Instrument spectral bound). *Let $s = \min\{d_A, d_B\}$ and $\vec{\Delta} \in \mathfrak{V}^{(n)}_{A,B}$. Let*

$$X = \widehat{J}_{\widehat{\vec{\Delta}}} \in \operatorname{Herm}(A_0 \otimes B \otimes C_n), \qquad X = \sum_{k=1}^{n} X_k \otimes |k\rangle\langle k|, \quad X_k = \widehat{J}_{\Delta_k},$$

be the normalised Choi operator of the flagged map $\widehat{\vec{\Delta}} : A \to B \otimes C_n$. Then

$$\max\{0, \lambda_{\max}(-X)\} \leq \frac{s}{2d_A} \|\vec{\Delta}\|_{\diamond,\text{inst}}.$$

Proof. If $\lambda_{\max}(-X) \leq 0$ there is nothing to prove. Otherwise, since X is block diagonal, $\lambda_{\max}(-X) = \max_k \lambda_{\max}(-X_k)$. Choose k_0 attaining the maximum and a unit vector $|v\rangle \in A_0 \otimes B$ with $X_{k_0}|v\rangle = -\lambda|v\rangle$, $\lambda = \lambda_{\max}(-X)$. Write $W \subseteq A_0$ for the support of the A_0-marginal $\operatorname{Tr}_B |v\rangle\langle v|$ and set $b = \dim W$; the marginal of a pure state has rank equal to its Schmidt rank, so $b \leq s = \min\{d_A, d_B\}$. If $|e_1\rangle, \ldots, |e_b\rangle$ is an orthonormal basis of W and $|\overline{e}_j\rangle \in A$ its coordinatewise conjugate under the fixed Choi bases, define

$$|\omega_W\rangle = \frac{1}{\sqrt{b}} \sum_{j=1}^{b} |e_j\rangle_{A_0} |\overline{e}_j\rangle_A.$$

With P_W the projection onto W, the normalised-Choi convention gives

$$Y := (\operatorname{id}_{A_0} \otimes \widehat{\vec{\Delta}})(|\omega_W\rangle\langle\omega_W|) = \frac{d_A}{b}(P_W \otimes I_{B \otimes C_n}) X (P_W \otimes I_{B \otimes C_n}).$$

The direction constraint implies

$$\operatorname{Tr} Y = \frac{d_A}{b} \operatorname{Tr}\left[P_W \sum_{k=1}^{n} \operatorname{Tr}_B X_k\right] = 0,$$

so Y is Hermitian and traceless. Since $|v\rangle \in W \otimes B$, testing on $|v, k_0\rangle := |v\rangle \otimes |k_0\rangle$ gives

$$\langle v, k_0|Y|v, k_0\rangle = -\frac{d_A}{b}\lambda.$$

A traceless Hermitian operator $Y = Y_+ - Y_-$ satisfies $\operatorname{Tr} Y_+ = \operatorname{Tr} Y_- = \frac{1}{2}\|Y\|_1$, and for every unit vector $|w\rangle$ one has $\langle w|Y|w\rangle \geq -\operatorname{Tr} Y_-$: indeed $Y \geq -Y_-$ implies $\langle w|Y|w\rangle \geq -\langle w|Y_-|w\rangle \geq -\operatorname{Tr} Y_-$. Consequently

$$\|Y\|_1 \geq -2\langle v, k_0|Y|v, k_0\rangle = \frac{2d_A}{b}\lambda \geq \frac{2d_A}{s}\lambda.$$

The state $|\omega_W\rangle\langle\omega_W|$ is admissible in the instrument diamond-norm optimisation, so $\|\vec{\Delta}\|_{\diamond,\mathrm{inst}} \geq \|Y\|_1$, and rearranging proves the claim. The entangled input thus plays two roles at once: its Schmidt support W selects the subspace on which the negative eigenvalue of the perturbation is read out, the normalised-Choi convention amplifying it by the factor d_A/b; and the same state is an admissible reference-assisted input, so the resulting trace-norm bound on Y is automatically a lower bound in the instrument diamond norm. This double role is the reason the input dimension appears in the denominator of the spectral bound: a perturbation can distribute its negativity across all d_A Choi coordinates, while the entangled input concentrates it back onto the b-dimensional Schmidt support of the witness, amplifying the negative eigenvalue by the factor d_A/b. □

Theorem 3.3 (Instrument diamond ball)**.** *Let $s = \min\{d_A, d_B\}$. Then*

$$\overline{B}^{\mathrm{rel}}_{\diamond,\mathrm{inst}}\left(\mathcal{I}_*, \frac{2}{nd_B s}\right) \subseteq \mathrm{Inst}_n(A, B).$$

Proof. Every eigenvalue of the flagged Choi operator of $\mathcal{I}_*$ equals $1/(nd_A d_B)$. If $\vec{\Delta} \in \mathfrak{V}^{(n)}_{A,B}$ with $\|\vec{\Delta}\|_{\diamond,\mathrm{inst}} \leq 2/(nd_B s)$ and $X = \widehat{J}_{\vec{\Delta}}$, Lemma 3.2 gives

$$\lambda_{\max}(-X) \leq \frac{s}{2d_A} \cdot \frac{2}{nd_B s} = \frac{1}{nd_A d_B} = \lambda_{\min}(\widehat{J}_{\mathcal{I}_*}),$$

so the full flagged Choi operator $\widehat{J}_{\mathcal{I}_*} + X$ is positive semidefinite. Since X is block diagonal this is equivalent to positivity of every outcome block, so each perturbed component is completely positive. The summed marginal is unchanged by the direction constraint, so the sum is trace preserving; and because the blocks are positive with $\sum_k \operatorname{Tr}_B(\widehat{J}_{(\mathcal{I}_*)_k} + X_k) = I_{A_0}/d_A$, each partial trace is bounded by I_{A_0}/d_A, so each component is trace-nonincreasing. Hence $\mathcal{I}_* + \vec{\Delta} \in \mathrm{Inst}_n(A, B)$. □

Theorem 3.4 (Exact centred instrument radius)**.** *Assume $nd_B > 1$, and let $s = \min\{d_A, d_B\}$. Then*

$$\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) = \frac{2}{nd_B s}.$$

Proof. Theorem 3.3 gives the lower bound. For the reverse inequality it suffices to exhibit a unit-norm direction reaching the boundary at the stated distance. The condition $nd_B > 1$ yields exactly two cases.

Case $d_B > 1$. Choose an s-dimensional input subspace $W_A \subseteq A$ and two isometries $V_0, V_1 : W_A \to B$ with $\operatorname{Tr}(V_0^\dagger V_1) = 0$ (if $s = 1$, two orthogonal output unit vectors; if $s \geq 2$, fix V_0, take a trace-zero unitary U on W_A, e.g. diagonal with roots of unity, and set $V_1 = V_0U$). With P_{W_A} the input projection define the completely positive, trace-nonincreasing maps

$$\mathcal{E}_j(Z) = V_j P_{W_A} Z P_{W_A} V_j^\dagger, \qquad j = 0, 1.$$

Choose an orthonormal basis $|e_1\rangle, \ldots, |e_s\rangle$ of W_A and set

$$|\omega_{W_A}\rangle = \frac{1}{\sqrt{s}} \sum_{a=1}^{s} |\overline{e}_a\rangle_{A_0} |e_a\rangle_A, \qquad P_j = (\mathrm{id} \otimes \mathcal{E}_j)(|\omega_{W_A}\rangle\langle\omega_{W_A}|).$$

The P_j are rank-one projectors with

$$\mathrm{Tr}_B\, P_0 = \mathrm{Tr}_B\, P_1 = \frac{P_{\overline{W}_A}}{s}, \qquad \mathrm{Tr}(P_0P_1) = \left|\frac{1}{s}\,\mathrm{Tr}(V_0^\dagger V_1)\right|^2 = 0,$$

so the maximally entangled input produces orthogonal pure outputs and $\|\mathcal{E}_1 - \mathcal{E}_0\|_\diamond = 2$. Define the instrument direction

$$\Delta_1 = \tfrac{1}{2}(\mathcal{E}_1 - \mathcal{E}_0), \qquad \Delta_k = 0 \ (k \geq 2).$$

Since $\mathrm{Tr}\,\mathcal{E}_1(Z) = \mathrm{Tr}(P_{W_A}Z) = \mathrm{Tr}\,\mathcal{E}_0(Z)$, the summed direction is trace-annihilating, so $\vec{\Delta} \in \mathfrak{V}^{(n)}_{A,B}$; its flagged diamond norm is 1 and its first normalised Choi block is $\widehat{J}_{\Delta_1} = \frac{s}{2d_A}(P_1 - P_0)$. At $t = 2/(nd_Bs)$ the first block of $\mathcal{I}_* + t\vec{\Delta}$ equals

$$\frac{1}{nd_Ad_B}\big(I_{A_0\otimes B} + P_1 - P_0\big),$$

which is positive semidefinite, with the unit vectors supporting P_0 as zero eigenvectors because $P_0P_1 = 0$. The remaining blocks are unchanged and the marginal constraint is preserved because $\mathrm{Tr}_B\, P_0 = \mathrm{Tr}_B\, P_1$. Hence the ray reaches the relative boundary at distance t. For $t' > t$ the eigenvalue on the support of P_0 becomes $1/(nd_Ad_B) - t's/(2d_A) < 0$, so $\mathcal{I}_* + t'\vec{\Delta}$ is not an instrument. No closed relative ball of radius larger than t is contained in $\mathrm{Inst}_n(A, B)$.

Case $d_B = 1$, $n \geq 2$ (POVMs). Choose a rank-one projection P on A and define the positive functional $\mathcal{F} : \mathrm{L}(A) \to \mathbb{C}$, $\mathcal{F}(Z) = \mathrm{Tr}(PZ)$. Define instrument directions as scalar-output maps

$$\Delta_1 = -\tfrac{1}{2}\mathcal{F}, \qquad \Delta_2 = +\tfrac{1}{2}\mathcal{F}, \qquad \Delta_k = 0 \ (k \geq 3),$$

so that $\Delta_1 + \Delta_2 = 0$ as linear functionals and the summed direction vanishes. For every reference-assisted input state ξ, set

$$\zeta_E = \mathrm{Tr}_A[(I_E \otimes P)\xi(I_E \otimes P)] = \mathrm{Tr}_A[(I_E \otimes P)\xi] \geq 0.$$

Then

$$(\mathrm{id}_E \otimes \widehat{\vec{\Delta}})(\xi) = \frac{1}{2}\,\zeta_E \otimes \big(|2\rangle\langle 2| - |1\rangle\langle 1|\big),$$

whose trace norm is $\mathrm{Tr}\,\zeta_E \leq 1$, with equality attained by an input supported on P; hence $\|\vec{\Delta}\|_{\diamond,\mathrm{inst}} = 1$. In the fixed Choi basis, $\widehat{J}_{\mathcal{F}} = P^\mathsf{T}/d_A$. At $t = 2/n$ the first two Choi blocks of $\mathcal{I}_* + t\vec{\Delta}$ are

$$\frac{I_{A_0} - P^\mathsf{T}}{nd_A}, \qquad \frac{I_{A_0} + P^\mathsf{T}}{nd_A},$$

all other blocks remaining $I_{A_0}/(nd_A)$. These blocks are positive semidefinite, the first is singular, and their sum is I_{A_0}/d_A; the ray reaches the POVM boundary at distance $2/n$, which is the stated formula for $d_B = s = 1$. For $t' > 2/n$ the first block has the negative eigenvalue $1/(nd_A) - t'/(2d_A)$ on the range of P^T, proving maximality. $\square$

Remark 3.5 (Physical origin of the radius)**.** *Each Choi block of the uniform instrument has positivity margin $1/(nd_Ad_B)$, while a unit instrument-diamond perturbation can create a negative Choi eigenvalue of size at most $s/(2d_A)$; their ratio is the exact radius $2/(nd_Bs)$. The factor $1/n$ reflects the distribution of the uniform trace budget among n outcomes* at the uniform centre*; by Theorem 3.12 no other centre carries a larger relative ball, so this value is also the global in-radius, but the specific value $2/(nd_Bs)$ and the $1/n$ mechanism are tied to the uniform centre and are not a universal $1/n$ law for arbitrary local robustness notions at other centres. The same numerical positivity-budget calculation is realised by two* distinct *witnesses living in different faces of the positive cone: an orthogonal-output isometry witness when $d_B > 1$, and an effect-transfer witness (a rank-one measurement effect moved between two classical outcomes) when $d_B = 1$, $n \geq 2$ (see Table 2).*

Remark 3.6 (Global in-radius)**.** *Theorem 3.4 determines the largest ball centred at $\mathcal{I}_*$. The stronger question — the global in-radius*

$$\sup_{\mathcal{I}_0 \in \mathrm{Inst}_n(A,B)} \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0)$$

over all centres — is settled by Theorem 3.12 (placed after Theorem 3.8, whose symmetrisation machinery it uses): the in-radius is a concave function of the centre, so the same Haar averaging that makes $\mathcal{I}_*$ optimal for the covering radius makes it optimal for the in-radius, and the supremum equals $\rho_n = 2/(nd_Bs)$.

Lemma 3.7 (Bipartite domination for instrument blocks)**.** *Let $\sigma_{EB} \geq 0$ be subnormalised, with $\rho_E = \mathrm{Tr}_B\, \sigma_{EB}$ and $r_E = \mathrm{rank}\, \rho_E$. Then*

$$\sigma_{EB} \leq \min\{r_E, d_B\}\, \rho_E \otimes I_B.$$

Proof. If $\rho_E = 0$ there is nothing to prove. Otherwise let $E_0 = \mathrm{supp}\, \rho_E$, so that $\rho_E^{-1/2}$ is defined on E_0 (extended by zero on the orthocomplement), and set

$$Z = (\rho_E^{-1/2} \otimes I_B)\sigma_{EB}(\rho_E^{-1/2} \otimes I_B).$$

Then $Z \geq 0$ and $\mathrm{Tr}_B\, Z = I_{E_0}$. Write $Z = \sum_\alpha |K_\alpha\rangle\!\rangle\langle\!\langle K_\alpha|$ with $\sum_\alpha K_\alpha^\dagger K_\alpha = I_{E_0}$ in the vectorisation convention $|K\rangle\!\rangle = \sum_i |\bar{i}\rangle \otimes K|i\rangle$, under which $\mathrm{Tr}_B\, |K\rangle\!\rangle\langle\!\langle K| = \overline{K^\dagger K}$, the entrywise conjugate: the constraint is unaffected because the identity is real, and the pairing $\langle\!\langle V|K\rangle\!\rangle = \mathrm{Tr}(V^\dagger K)$ is the one used below. For a unit vector $|v\rangle \in E_0 \otimes B$ let $V : E_0 \to B$ be the corresponding matrix of rank $b \leq \min\{r_E, d_B\}$. Using $|\mathrm{Tr}\, M|^2 \leq \mathrm{rank}(M)\|M\|_2^2$ with $M = K_\alpha V^\dagger$ of rank at most b,

$$\langle v|Z|v\rangle = \sum_\alpha |\mathrm{Tr}(V^\dagger K_\alpha)|^2 \leq b \sum_\alpha \mathrm{Tr}(V K_\alpha^\dagger K_\alpha V^\dagger) = b.$$

Thus $Z \leq \min\{r_E, d_B\} I_{E_0 \otimes B}$, and conjugation by $\rho_E^{1/2} \otimes I_B$ proves the claim. □

Theorem 3.8 (Exact instrument covering radius)**.** *Let $s = \min\{d_A, d_B\}$. For $\Psi \in \mathfrak{A}_{A,B}^{(n)}$ define*

$$R_{\mathrm{inst}}(\Psi) = \sup_{\mathcal{I} \in \mathrm{Inst}_n(A,B)} \|\mathcal{I} - \Psi\|_{\diamond,\mathrm{inst}}.$$

Then the uniform depolarising instrument is an optimal centre over all of $\mathfrak{A}_{A,B}^{(n)}$, and

$$R_{\mathrm{inst}}(\mathcal{I}_*) = 2\Big(1 - \frac{1}{nd_Bs}\Big) = 2 - \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*).$$

Consequently the exact constant-code instrument width equals this value. For $n = 1$ this reduces to the exact Chebyshev covering radius for channels established in Ref. [2].

Proof. Fix a reference-assisted input state, taking the reference of input dimension d_A (sufficient for the diamond norm, as recorded in the definition of the instrument diamond norm), and write the flagged output of an instrument as

$$\sigma_{EBC_n} = \sum_{k=1}^{n} \sigma_{k,EB} \otimes |k\rangle\langle k|, \qquad \sum_k \mathrm{Tr}_B\, \sigma_{k,EB} = \rho_E.$$

Put $\rho_{k,E} = \mathrm{Tr}_B\, \sigma_{k,EB}$. Since $\mathrm{rank}\, \rho_{k,E} \le d_A$, Lemma 3.7 gives

$$\sigma_{k,EB} \le s\, \rho_{k,E} \otimes I_B \le s\, \rho_E \otimes I_B.$$

Therefore

$$\sigma_{EBC_n} \le n d_B s \left(\rho_E \otimes \frac{I_B}{d_B} \otimes \frac{I_{C_n}}{n} \right),$$

and the state in parentheses is the output of $\mathcal{I}_*$. For states with $\sigma \le K\tau$, writing P for the positive spectral projection of $\sigma - \tau$, the quantities $a = \mathrm{Tr}(P\sigma)$ and $b = \mathrm{Tr}(P\tau)$ satisfy $a \le Kb$ (as $P\sigma P \le KP\tau P$) and $a \le 1$, whence

$$\tfrac{1}{2}\|\sigma - \tau\|_1 = a - b \le a - \frac{a}{K} \le 1 - \frac{1}{K}.$$

Taking the supremum over inputs yields $R_{\mathrm{inst}}(\mathcal{I}_*) \le 2(1 - 1/(n d_B s))$.

For the matching lower bound, select an s-dimensional subspace W of the input space, an isometry $V : W \to B$, and an output state ω. With P_W the projection onto W, the channel

$$\Phi(X) = V P_W X P_W V^\dagger + \mathrm{Tr}[(I_A - P_W)X]\, \omega$$

acts as a pure isometry on inputs supported on W and as a fixed replacement off W. Place it in the first outcome, $\mathcal{I}_1 = \Phi$, $\mathcal{I}_k = 0$ for $k \ge 2$. On a maximally entangled input of Schmidt rank s supported on W, the flagged output of $\mathcal{I}$ is pure, whereas the uniform instrument emits the maximally mixed state on the $n d_B s$-dimensional support of its own output; the trace norm of the difference of the two flagged outputs is $2(1 - 1/(n d_B s))$ (trace distance $1 - 1/(n d_B s)$). Hence $R_{\mathrm{inst}}(\mathcal{I}_*) \ge 2(1 - 1/(n d_B s))$, and equality holds.

Optimality of the centre. Let $G = \mathrm{U}(B) \times S_n$ act on $\mathfrak{A}^{(n)}_{A,B}$ by

$$[(U, \pi) \cdot \Psi]_k = \mathcal{U}_U \circ \Psi_{\pi^{-1}(k)}, \qquad \mathcal{U}_U(X) = U X U^\dagger.$$

This action preserves $\mathrm{Inst}_n(A, B)$ and the instrument diamond metric, hence $R_{\mathrm{inst}}(g \cdot \Psi) = R_{\mathrm{inst}}(\Psi)$. For each fixed $\mathcal{I}$, the function $\Psi \mapsto \|\mathcal{I} - \Psi\|_{\diamond,\mathrm{inst}}$ is convex (a norm composed with a translation), and a supremum of convex functions is convex, so the functional R_{inst} is convex. Let $\overline{\Psi}$ be the average of Ψ over G with respect to Haar measure on $\mathrm{U}(B)$ and the uniform measure on S_n. Jensen's inequality gives $R_{\mathrm{inst}}(\overline{\Psi}) \le R_{\mathrm{inst}}(\Psi)$. Averaging first over $\mathrm{U}(B)$, Schur's lemma [12] gives for each component

$$\int_{\mathrm{U}(B)} U \Psi_k(X) U^\dagger dU = \frac{\mathrm{Tr}(\Psi_k(X))}{d_B} I_B;$$

averaging then over outcome permutations, the k-th component becomes

$$\frac{1}{n} \sum_{j=1}^{n} \frac{\mathrm{Tr}(\Psi_j(X))}{d_B} I_B = \frac{\mathrm{Tr}(X)}{n d_B} I_B,$$

because the summed map is trace preserving. Thus $\overline{\Psi} = \mathcal{I}_*$, and $R_{\mathrm{inst}}(\mathcal{I}_*) \le R_{\mathrm{inst}}(\Psi)$ for every $\Psi \in \mathfrak{A}^{(n)}_{A,B}$. Since $\mathcal{I}_* \in \mathrm{Inst}_n(A, B)$, the constant-code width equals the covering radius about $\mathcal{I}_*$. $\square$

Proposition 3.9 (Exact instrument diameter)**.** *Assume $nd_B > 1$. Then*

$$\operatorname{diam}_{\diamond,\mathrm{inst}}(\mathrm{Inst}_n(A,B)) = 2.$$

Proof. For any two instruments $\mathcal{I}, \mathcal{J}$, the flagged maps $\widehat{\mathcal{I}}, \widehat{\mathcal{J}}$ are channels, so

$$\|\mathcal{I} - \mathcal{J}\|_{\diamond,\mathrm{inst}} = \|\widehat{\mathcal{I}} - \widehat{\mathcal{J}}\|_\diamond \le \|\widehat{\mathcal{I}}\|_\diamond + \|\widehat{\mathcal{J}}\|_\diamond = 2.$$

Equality is achieved by replacement instruments with orthogonal flagged states, which exist whenever $nd_B > 1$: if $d_B \ge 2$, take two orthogonal output states in the same flag; if $d_B = 1$, then $n \ge 2$, and take two orthogonal flags. Testing the difference of the two replacement instruments on any input state whose flagged output is supported in the flag pair produces two orthogonally supported flagged states (pure outputs when the input is a product pure state), at trace distance 2, so the instrument diamond distance is at least 2; with the upper bound above it is exactly 2. □

Corollary 3.10 (Metric complementarity of the uniform instrument)**.** *Assume $nd_B > 1$. Then*

$$\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) + R_{\mathrm{inst}}(\mathcal{I}_*) = 2 = \operatorname{diam}_{\diamond,\mathrm{inst}}(\mathrm{Inst}_n(A,B)).$$

This is a numerical complementarity identity for the centred in-radius and the covering radius about $\mathcal{I}_*$: it is obtained by adding the two exact values, each witnessed by its own, independently constructed point of the body.

Proof. By Theorems 3.4 and 3.8,

$$\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) + R_{\mathrm{inst}}(\mathcal{I}_*) = \frac{2}{nd_Bs} + 2\Big(1 - \frac{1}{nd_Bs}\Big) = 2,$$

and $\operatorname{diam}_{\diamond,\mathrm{inst}}(\mathrm{Inst}_n(A,B)) = 2$ by Proposition 3.9. The identity is purely algebraic: the two summands are witnessed by different, independently constructed points of the body. □

Lemma 3.11 (Concavity and Lipschitz continuity of the in-radius)**.** *Write*

$$B_{\mathfrak{V}}(\rho) = \big\{\vec{\Delta} \in \mathfrak{V}_{A,B}^{(n)} : \|\vec{\Delta}\|_{\diamond,\mathrm{inst}} \le \rho\big\}$$

for the closed direction ball of radius ρ. Assume $nd_B > 1$, excluding the singleton case of Remark 2.5. For centres $\mathcal{I}_0, \mathcal{I}_1 \in \mathrm{Inst}_n(A,B)$ and $\lambda \in [0,1]$,

$$\rho_{\diamond,\mathrm{inst}}\big(\lambda\mathcal{I}_0 + (1-\lambda)\mathcal{I}_1\big) \ge \lambda\,\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) + (1-\lambda)\,\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_1),$$

and

$$\big|\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) - \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_1)\big| \le \|\mathcal{I}_0 - \mathcal{I}_1\|_{\diamond,\mathrm{inst}}.$$

In particular, the in-radius is a continuous concave function of the centre on the body.

Proof. For centrally symmetric balls of one fixed norm one has the Minkowski identity

$$\lambda B_{\mathfrak{V}}(\rho_0) + (1-\lambda)B_{\mathfrak{V}}(\rho_1) = B_{\mathfrak{V}}\big(\lambda\rho_0 + (1-\lambda)\rho_1\big) :$$

the inclusion "$\subseteq$" is the triangle inequality, and "$\supseteq$" follows from the decomposition $\rho u = \lambda\rho_0 u + (1-\lambda)\rho_1 u$ with $\rho = \lambda\rho_0 + (1-\lambda)\rho_1$ and $\|u\|_{\diamond,\mathrm{inst}} \le 1$. Let $\rho_i < \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_i)$ be admissible radii, i.e. $\mathcal{I}_i + B_{\mathfrak{V}}(\rho_i) \subseteq \mathrm{Inst}_n(A,B)$ for $i = 0,1$ (admissible radii exist below the supremum, which need not itself be attained); every point of $\lambda\mathcal{I}_0 + (1-\lambda)\mathcal{I}_1 + \lambda B_{\mathfrak{V}}(\rho_0) + (1-\lambda)B_{\mathfrak{V}}(\rho_1)$ is a convex combination of two instruments, hence an instrument. Taking the supremum over admissible ρ_0 and ρ_1, which factorises because the two containment conditions are independent of each other, gives the concavity.

For the Lipschitz bound, let $\delta = \|\mathcal{I}_0 - \mathcal{I}_1\|_{\diamond,\mathrm{inst}}$, so $\mathcal{I}_1 - \mathcal{I}_0 \in B_{\mathfrak{V}}(\delta)$. If $\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) \le \delta$ there is nothing to prove. Otherwise, with $\rho = \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0)$,

$$\mathcal{I}_1 + B_{\mathfrak{V}}(\rho-\delta) = \mathcal{I}_0 + (\mathcal{I}_1 - \mathcal{I}_0) + B_{\mathfrak{V}}(\rho-\delta) \subseteq \mathcal{I}_0 + B_{\mathfrak{V}}(\delta) + B_{\mathfrak{V}}(\rho-\delta) = \mathcal{I}_0 + B_{\mathfrak{V}}(\rho) \subseteq \mathrm{Inst}_n(A,B),$$

so $\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_1) \ge \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) - \delta$; exchanging the two centres gives the absolute value. □

Theorem 3.12 (Global in-radius)**.** *Assume $nd_B > 1$. Then the uniform depolarising instrument is an optimal in-radius centre, and*

$$\sup_{\mathcal{I}_0 \in \mathrm{Inst}_n(A,B)} \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) = \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) = \frac{2}{nd_B \min\{d_A, d_B\}}.$$

For $n = 1$ this is the global in-radius statement for channels of Ref. [2].

Proof. Let $G = \mathrm{U}(B) \times S_n$ act on $\mathfrak{A}^{(n)}_{A,B}$ as in the proof of Theorem 3.8, and let μ be the normalised Haar measure. The action restricts to the direction space: for $\vec{\Delta} \in \mathfrak{V}^{(n)}_{A,B}$, the components of $g \cdot \vec{\Delta}$ are $\mathcal{U}_U \circ \Delta_{\pi^{-1}(k)}$, and the A_0-marginal of a Choi operator is invariant under unitary conjugation of B,

$$\mathrm{Tr}_B\, \widehat{J}_{\mathcal{U}_U \circ \Delta} = \mathrm{Tr}_B \left[(I_{A_0} \otimes U) \widehat{J}_\Delta (I_{A_0} \otimes U^\dagger) \right] = \mathrm{Tr}_B\, \widehat{J}_\Delta,$$

so the summed marginal of $g \cdot \vec{\Delta}$ still vanishes. With W_π the flag-permutation unitary on C_n ($W_\pi |j\rangle = |\pi(j)\rangle$), the flagged operator of $g \cdot \vec{\Delta}$ satisfies

$$\widehat{g \cdot \vec{\Delta}}(X) = (U \otimes W_\pi)\, \widehat{\vec{\Delta}}(X)\, (U^\dagger \otimes W_\pi^\dagger) \qquad (X \in \mathrm{L}(A)),$$

that is, $\widehat{g \cdot \vec{\Delta}} = \mathcal{U}_{U \otimes W_\pi} \circ \widehat{\vec{\Delta}}$ is the post-composition of $\widehat{\vec{\Delta}}$ by the unitary channel of $U \otimes W_\pi$ on $B \otimes C_n$; post-composition by unitary channels preserves the diamond norm. Hence g maps $B_{\mathfrak{V}}(\rho)$ onto itself and $\mathrm{Inst}_n(A, B)$ onto itself, and consequently

$$\rho_{\diamond,\mathrm{inst}}(g \cdot \mathcal{I}_0) = \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0) \qquad (g \in G,\ \mathcal{I}_0 \in \mathrm{Inst}_n(A, B)).$$

By the Schur-lemma and permutation averages computed in the proof of Theorem 3.8 — valid verbatim for every point of $\mathfrak{A}^{(n)}_{A,B}$ — the barycentre of every orbit is $\mathcal{I}_*$:

$$\int_G g \cdot \mathcal{I}_0 \, d\mu(g) = \mathcal{I}_*.$$

The orbit map $g \mapsto g \cdot \mathcal{I}_0$ is continuous, and the in-radius is concave and continuous on the body (Lemma 3.11); Jensen's inequality (the integral being a limit of convex combinations of orbit points, to which concavity extends by continuity) gives

$$\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) = \rho_{\diamond,\mathrm{inst}}\left(\int_G g \cdot \mathcal{I}_0 \, d\mu(g) \right) \geq \int_G \rho_{\diamond,\mathrm{inst}}(g \cdot \mathcal{I}_0)\, d\mu(g) = \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0).$$

Together with the exact centred value of Theorem 3.4, this proves the claim. ☐

Remark 3.13 (Concave–convex duality of the two radii)**.** *Theorem 3.12 is the mirror image of the covering-radius symmetrisation in Theorem 3.8. The covering radius is a convex function of the centre, and Jensen's inequality at the Haar barycentre $\mathcal{I}_*$ shows that no centre has a smaller covering radius; the in-radius is a concave function of the centre (Lemma 3.11), and the same Jensen inequality shows that no centre has a larger in-radius. The uniform depolarising instrument is thus an optimal centre for both radii simultaneously, and the complementarity identity of Corollary 3.10 holds at this common optimal centre. Two consequences follow. First, the ball term $\rho_n = 2/(nd_B s)$ of the certified intermediate lower bound $\gamma^{(n)}_{A,B} = \max\{\rho_n, 1/d_A\}$ is optimal among all centred-ball certificates: re-centring the radial construction cannot certify more than ρ_n at any centre. Second, the Lipschitz bound of Lemma 3.11 is a robustness statement: every centre within distance ε of $\mathcal{I}_*$ still carries a relative ball of radius $\rho_n - \varepsilon$. Uniqueness of the optimal centre is left open.*

4 Antipodal certificates and the lower envelope

The certified lower bounds rest on a single topological device, the antipodal profile of the body: an antipodally separated sphere inside the body forces every compression with that many latent dimensions to err. In operational terms, each such sphere is a compact family of instruments in which every member is paired with an antipode at large instrument-diamond distance (maximal, 2, for the replacement, observable, and joined constructions; $2/d_A$ for the full-direction sphere), and Borsuk–Ulam forces some paired members to share a latent code, so no reconstruction can serve both without error. This section builds three such spheres — the replacement sphere, the full-direction sphere, and the observable sphere — and combines them into the certified lower envelope $\beta_r^{\mathrm{best},(n)}$; the first and the third are combined by an instrument-level join (Proposition 4.12).

Throughout this section, instruments are identified with their normalised flagged Choi operators, a compact convex subset of a finite-dimensional real vector space. All constructed maps are finite-dimensional and continuous in Choi trace norm or operator norm; by norm equivalence on this finite-dimensional space, this is equivalent to continuity in the instrument diamond norm, which is the metric used in the definition of the antipodal profile. The two notions of continuity are therefore used interchangeably.

Definition 4.1 (Antipodal profile). *For any nonempty metric space (K, d) define*

$$\alpha_r(K) = \frac{1}{2} \sup_{h \in C(S^r, K)} \min_{u \in S^r} d(h(u), h(-u)),$$

with the instrument diamond metric when $K \subseteq \mathrm{Inst}_n(A, B)$. The same quantity is defined for arbitrary compact metric sources in Ref. [2], where its general properties (monotonicity, behaviour under subsets and Lipschitz maps, metric perturbations) are recorded.

Remark 4.2 (Terminology for the four radius notions). *Four distinct radius-like quantities are used throughout this paper, always written with their qualifiers: (i) the* centred relative in-radius $\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0)$*, henceforth abbreviated to the centred in-radius, the largest relative ball about a fixed centre that is contained in the body; (ii) the* global in-radius $\sup_{\mathcal{I}_0} \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_0)$*, which by Theorem 3.12 equals the centred in-radius at $\mathcal{I}_*$; (iii) the* worst-case covering radius $R_{\mathrm{inst}}(\Psi)$*, the maximal distance from a centre to the body; and (iv) the* antipodal profile α_r*, a width-like sequence of sphere index r, which is not the radius of any ball. In what follows the unqualified word "radius" is not used. Figure 1 sketches the four notions.*

Proposition 4.3 (Monotonicity of profiles, widths, and envelopes). *For every nonempty metric space K,*

$$\alpha_{r+1}(K) \le \alpha_r(K) \qquad (r \ge 0),$$

and if K is compact then $0 \le \alpha_r(K) \le \frac{1}{2}\operatorname{diam}(K)$; consequently, if $\alpha_D(K) = 0$ then $\alpha_r(K) = 0$ for all $r \ge D$. For the instrument body,

$$\delta^{\diamond}_{r+1,\mathrm{inst}}(A, B) \le \delta^{\diamond}_{r,\mathrm{inst}}(A, B),$$

and the certified envelopes (Definitions 4.17 and 5.14 below) satisfy

$$\beta_{r+1}^{\mathrm{best},(n)}(A, B) \le \beta_r^{\mathrm{best},(n)}(A, B), \qquad U_{r+1}^{(n)}(A, B) \le U_r^{(n)}(A, B).$$

Proof. For the antipodal profile, let $h : S^{r+1} \to K$ be continuous and restrict it to an equatorial copy $S^r \simeq \{(x_0, \ldots, x_{r+1}) \in S^{r+1} : x_{r+1} = 0\}$. The antipodal map on the equator is the ordinary antipodal map on S^r, hence

$$\min_{u \in S^r} d(h(u), h(-u)) \ge \min_{u \in S^{r+1}} d(h(u), h(-u)).$$

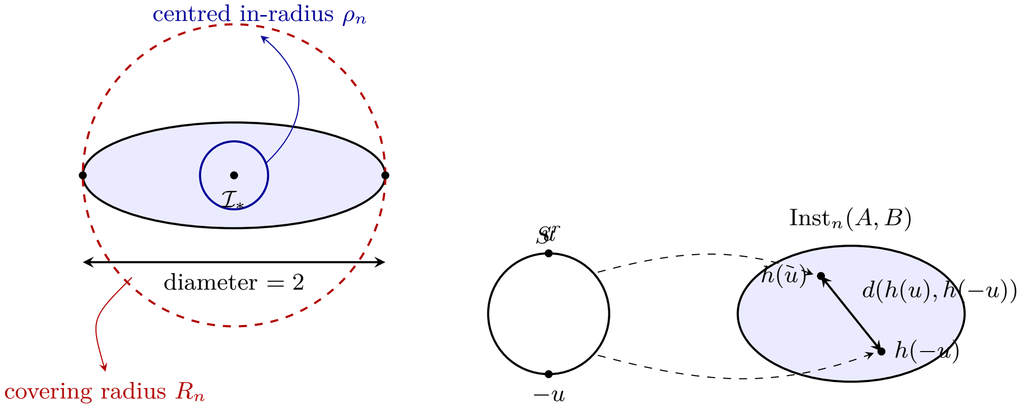

Figure 1: The four radius notions of Remark 4.2. Left: the centred in-radius ρ_n (largest ball about $\mathcal{I}_*$ contained in the body), the covering radius R_n (smallest ball about $\mathcal{I}_*$ containing the body), and the diameter, attained by two points of the body at distance 2. Right: the antipodal profile: for a continuous map $h : S^r \to \mathrm{Inst}_n(A, B)$, α_r is, up to the factor 1/2, the largest value of $\min_u d(h(u), h(-u))$ that can be forced; it is governed by antipodal separation, not by balls. The drawing is schematic; the numerical identity $\rho_n + R_n = 2$ of Corollary 3.10 holds for the exact values of the instrument body, not for a generic convex set.

Taking one half and the supremum over all continuous $h : S^{r+1} \to K$ gives $\alpha_r(K) \geq \alpha_{r+1}(K)$. Distances are nonnegative, and $d(h(u), h(-u)) \leq \mathrm{diam}(K)$ for every h and u, so $0 \leq \alpha_r(K) \leq \frac{1}{2}\,\mathrm{diam}(K)$. If $\alpha_D(K) = 0$, the inequality $\alpha_r(K) \leq \alpha_D(K) = 0$ for $r \geq D$ follows by induction from $\alpha_{r+1} \leq \alpha_r$, and $\alpha_r(K) \geq 0$ always, so $\alpha_r(K) = 0$ for all $r \geq D$.

For the widths, fix any continuous encoder $f : \mathrm{Inst}_n(A, B) \to \mathbb{R}^r$ and any decoder $g : \mathbb{R}^r \to \mathrm{Inst}_n(A, B)$, and define $f' : \mathrm{Inst}_n(A, B) \to \mathbb{R}^{r+1}$ by $f'(x) = (f(x), 0)$ and $g' : \mathbb{R}^{r+1} \to \mathrm{Inst}_n(A, B)$ by $g'(z_1, \ldots, z_{r+1}) = g(z_1, \ldots, z_r)$. Then $g' \circ f' = g \circ f$ pointwise, so the pair (f', g') achieves the same error at latent dimension $r+1$ that (f, g) achieves at dimension r; taking the infimum over (f, g) gives $\delta^{\diamond}_{r+1,\mathrm{inst}}(A, B) \leq \delta^{\diamond}_{r,\mathrm{inst}}(A, B)$.

For the lower envelope, the piecewise definition gives $\beta_0^{\mathrm{best},(n)} = R_n$, $\beta_r^{\mathrm{best},(n)} = 1$ on $1 \leq r \leq L^{(n)}_{A,B}$, $\gamma^{(n)}_{A,B}$ on $L^{(n)}_{A,B} < r < D^{(n)}_{A,B}$, and 0 for $r \geq D^{(n)}_{A,B}$; since $nd_B s \geq 2$ gives $R_n \geq 1$ and $0 < \gamma^{(n)}_{A,B} \leq 1$ (Definition 4.17), the sequence is non-increasing.

For the upper envelope, each of the four families of admissible terms in Definition 5.14 is monotone in r: the constant-code term is independent of r; the convex-projection term $2(D^{(n)}_{A,B} - r)/(D^{(n)}_{A,B} - r + 1)$ is decreasing in r; and the pinching families are defined by threshold conditions $r \geq r^{(n)}_{\mathrm{out}}$ and $r \geq r^{(n)}_{\mathrm{in}}$, so every pinching term admissible at r remains admissible at $r+1$, while further terms may become admissible. Hence every term entering $U^{(n)}_r$ is bounded below by a term entering $U^{(n)}_{r+1}$ (namely the same term, or a smaller newly available one), and $U^{(n)}_{r+1}(A, B) \leq U^{(n)}_r(A, B)$. $\square$

Let

$$\mathcal{S}_{\mathrm{flag}}(B \otimes C_n) = \Big\{ \tau = \sum_{k=1}^{n} \tau_k \otimes |k\rangle\langle k| : \tau_k \geq 0,\ \sum_{k=1}^{n} \mathrm{Tr}\,\tau_k = 1 \Big\}.$$

For $\tau \in \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ define the replacement instrument $\vec{\mathcal{R}}_\tau$ by $(\vec{\mathcal{R}}_\tau)_k(X) = \mathrm{Tr}(X)\tau_k$; each component is completely positive and trace-nonincreasing and the sum is a channel. Write

$$\mathrm{RepInst}_n(A, B) = \{\vec{\mathcal{R}}_\tau : \tau \in \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)\}.$$

Lemma 4.4 (Replacement-instrument isometry). *For all flagged states τ, σ,*

$$\|\vec{\mathcal{R}}_\tau - \vec{\mathcal{R}}_\sigma\|_{\diamond,\mathrm{inst}} = \|\tau - \sigma\|_1 = \sum_{k=1}^{n} \|\tau_k - \sigma_k\|_1.$$

Proof. The flagged channel of $\vec{\mathcal{R}}_\tau$ is the replacement channel $X \mapsto \mathrm{Tr}(X)\tau$. For every reference-assisted input state ξ,

$$(\mathrm{id} \otimes (\widehat{\vec{\mathcal{R}}_\tau} - \widehat{\vec{\mathcal{R}}_\sigma}))(\xi) = \mathrm{Tr}_A(\xi) \otimes (\tau - \sigma),$$

and $\mathrm{Tr}_A(\xi)$ is a state, so the trace norm equals $\|\tau - \sigma\|_1$ by multiplicativity; the supremum over inputs is unnecessary. The second equality follows from block diagonality in the flag. □

Theorem 4.5 (Perfectly separated replacement instruments). *Assume $nd_B > 1$. Replacement instruments contain a continuous image of $S^{nd_B^2-2}$ whose antipodes have instrument-diamond distance two, and consequently*

$$\alpha_r(\mathrm{Inst}_n(A, B)) = 1 \qquad (0 \le r \le nd_B^2 - 2).$$

For the replacement family itself the profile is exact:

$$\alpha_r(\mathrm{RepInst}_n(A, B)) = \begin{cases} 1, & 0 \le r \le nd_B^2 - 2, \\ 0, & r \ge nd_B^2 - 1. \end{cases}$$

Proof. The real vector space of flag-block-diagonal Hermitian operators on $B \otimes C_n$ has dimension nd_B^2; its trace-zero subspace

$$\mathfrak{H}_{\mathrm{flag},0} = \Big\{ X = \sum_{k=1}^{n} X_k \otimes |k\rangle\langle k| : X_k \in \mathrm{Herm}(B),\ \sum_k \mathrm{Tr}\, X_k = 0 \Big\}$$

has dimension $m := nd_B^2 - 1$. Choose any Euclidean norm and let $S(\mathfrak{H}_{\mathrm{flag},0}) \cong S^{m-1} = S^{nd_B^2-2}$ be its unit sphere. For X on the sphere write the Jordan decomposition $X = X_+ - X_-$; functional calculus preserves block diagonality and makes $X \mapsto X_+$ continuous. Since X is nonzero and traceless, $t_X := \mathrm{Tr}\, X_+ = \mathrm{Tr}\, X_- > 0$. The map $h(X) = X_+/t_X$ is a continuous map into $\mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ — a continuous image of the sphere, and injectivity, which this map may lack, is not required by the antipodal-width arguments — and $h(-X) = X_-/t_X$ has orthogonal support, so $\|h(X) - h(-X)\|_1 = 2$. Composing with the replacement embedding of Lemma 4.4 gives the claimed sphere of instruments; equatorial restriction supplies every smaller index, and the reverse inequality follows from the diameter bound (Proposition 3.9).

For the replacement family above this range, Lemma 4.4 identifies $\mathrm{RepInst}_n(A, B)$ affinely and isometrically with the compact convex body of flagged states inside a real affine space of dimension $m = nd_B^2 - 1$; choose an affine isomorphism of its affine hull with $\mathbb{R}^m$ and compose any continuous $h : S^r \to \mathrm{RepInst}_n(A, B)$ with it. For $r = m$, Borsuk–Ulam [15] applied to $S^m \to \mathbb{R}^m$ produces an antipodal collision; for $r > m$, restrict the map to an equatorial $S^m \subset S^r$ first. Every such map therefore has minimum antipodal distance zero, proving the second part of the formula. □

Remark 4.6 (In-radius versus antipodal profile). *$\rho_n = 2/(nd_B s)$ certifies a centrally symmetric norm ball in the full direction space $\mathfrak{V}_{A,B}^{(n)}$, while the sphere $S^{nd_B^2-2}$ via $X \mapsto X_+/\,\mathrm{Tr}\, X_+$ certifies a non-linear, antipodally 2-separated sphere in the replacement body. There is no contradiction between $\rho_n < 1$ and $\alpha_r = 1$ for $r \le nd_B^2 - 2$: these are different geometric mechanisms (a ball radius versus an antipodal profile) acting on different subsets of the body. The ball bound governs the intermediate regime, while the replacement sphere governs the small-index regime.*

Theorem 4.7 (Full-direction antipodal sphere)**.** *Assume $nd_B > 1$, and let $D = D^{(n)}_{A,B}$. For every $0 \le r \le D-1$ there exists a continuous map $h : S^r \to \mathrm{Inst}_n(A,B)$ such that*

$$\|h(u) - h(-u)\|_{\diamond,\mathrm{inst}} \ge \frac{2}{d_A} \qquad \forall u \in S^r.$$

Consequently $\alpha_r(\mathrm{Inst}_n(A,B)) \ge 1/d_A$ for all $0 \le r < D$. When $d_A = 1$ the separation is 2, hence $\alpha_r = 1$ for all $r < D^{(n)}_{1,B}$, consistent with the exact replacement profile. For $n = 1$ the construction restricts to the full-direction sphere of Ref. [2].

Proof. Identify instruments with their normalised flagged Choi operators

$$J \in \mathrm{Herm}(A_0 \otimes B \otimes C_n), \qquad J \ge 0, \qquad \mathrm{Tr}_{B\otimes C_n} J = \frac{I_{A_0}}{d_A},$$

flag-block-diagonal in C_n. The direction space

$$\mathcal{V} = \big\{Z \in \mathrm{Herm}(A_0 \otimes B \otimes C_n)_{\mathrm{flag}} : \mathrm{Tr}_{B\otimes C_n} Z = 0\big\}$$

has real dimension D. Equip $\mathcal{V}$ with a Euclidean norm and let $S(\mathcal{V}) \cong S^{D-1}$ be its unit sphere; it suffices to construct the map on S^{D-1}, since equatorial restriction supplies every smaller r. Fix the uniform flagged state

$$\tau_0 = \frac{I_B}{d_B} \otimes \frac{I_{C_n}}{n}.$$

For $Z \in S(\mathcal{V})$ write the Jordan decomposition $Z = Z_+ - Z_-$ with $Z_+, Z_- \ge 0$, $Z_+Z_- = 0$; functional calculus preserves flag-block-diagonality. Since $\mathrm{Tr}_{B\otimes C_n} Z = 0$,

$$\mathrm{Tr}_{B\otimes C_n} Z_+ = \mathrm{Tr}_{B\otimes C_n} Z_-.$$

Let $t = \mathrm{Tr}\, Z_+ = \mathrm{Tr}\, Z_-$. We claim $t > 0$: if $t = 0$ then $Z_+ = 0$, so $Z = -Z_- \le 0$; but then $\mathrm{Tr}\, Z = \mathrm{Tr}_{A_0} \mathrm{Tr}_{B\otimes C_n} Z = 0$ with $Z \le 0$ forces $Z = 0$, contradicting $Z \in S(\mathcal{V})$. Define the normalised states

$$\Theta_+ = \frac{Z_+}{t}, \qquad \Theta_- = \frac{Z_-}{t},$$

which have orthogonal supports and share the same A_0-marginal

$$\mu := \mathrm{Tr}_{B\otimes C_n} \Theta_+ = \mathrm{Tr}_{B\otimes C_n} \Theta_-.$$

If $d_A = 1$ the marginal condition is automatic (A_0 is one-dimensional and $\mu = 1$), and we set $J^+(Z) = \Theta_+$, $J^-(Z) = \Theta_-$. Otherwise put $\lambda = 1/d_A$. Since μ is a state on A_0, $\mu \le I_{A_0}$, hence $\lambda\mu \le I_{A_0}/d_A$. Define the correction state

$$\nu = \frac{\frac{I_{A_0}}{d_A} - \lambda\mu}{1-\lambda} = \frac{I_{A_0} - \mu}{d_A - 1} \ge 0, \qquad \mathrm{Tr}\,\nu = 1,$$

and set

$$J^+(Z) = \lambda\Theta_+ + (1-\lambda)\nu \otimes \tau_0, \qquad J^-(Z) = \lambda\Theta_- + (1-\lambda)\nu \otimes \tau_0.$$

Both $J^\pm(Z)$ are positive and flag-block-diagonal: the $\Theta_\pm$ summands inherit this property from Z through the Jordan decomposition, and the correction term $\nu\otimes\tau_0$ is flag-block-diagonal because τ_0 is diagonal in the fixed flag basis. Their common A_0-marginal is

$$\lambda\mu + (1-\lambda)\nu = \frac{I_{A_0}}{d_A},$$

so each is the normalised flagged Choi operator of an instrument; let $h(Z)$ be the instrument with Choi operator $J^+(Z)$. For $-Z$ the positive and negative parts are interchanged while t and

μ are unchanged, so ν is unchanged and $h(-Z)$ is the instrument with Choi operator $J^-(Z)$. The Choi difference is

$$J^+(Z) - J^-(Z) = \lambda(\Theta_+ - \Theta_-)$$

(with $\lambda = 1$ when $d_A = 1$), and since Θ_+, Θ_- are states with orthogonal supports,

$$\|J^+(Z) - J^-(Z)\|_1 = 2\lambda = \frac{2}{d_A}.$$

For any Hermiticity-preserving tuple $\vec{\Delta}$,

$$\|\vec{\Delta}\|_{\diamond,\text{inst}} \geq \|\widehat{J}_{\widehat{\Delta}}\|_1,$$

because testing the flagged map on the maximally entangled input state produces its normalised Choi operator. Therefore

$$\|h(Z) - h(-Z)\|_{\diamond,\text{inst}} \geq \frac{2}{d_A}.$$

Finally, $Z \mapsto Z_+$ is continuous on Hermitian operators and $t = \operatorname{Tr} Z_+$ is continuous and strictly positive on the compact unit sphere $S(\mathcal{V})$, hence bounded away from zero there; so $Z \mapsto J^+(Z)$ is continuous. Equatorial restriction proves the theorem for all $0 \leq r \leq D - 1$. □

Remark 4.8 (Tunable mixing parameter)**.** *In the proof of Theorem 4.7, the mixing parameter may be chosen as a function of Z: since $\mu = \mu(Z)$ is a state on A_0 we have $\|\mu\|_\infty \in [1/d_A, 1]$, and $I_{A_0}/d_A - \lambda\mu \geq 0$ holds if and only if $\lambda \leq 1/(d_A\|\mu\|_\infty)$. Taking $\lambda(Z) = 1/(d_A\|\mu(Z)\|_\infty) \in [1/d_A, 1]$ therefore increases the pointwise separation to*

$$\|J^+(Z) - J^-(Z)\|_1 = \frac{2}{d_A\|\mu(Z)\|_\infty} \geq \frac{2}{d_A},$$

with $\lambda = 1$ exactly when $\mu(Z) = I_{A_0}/d_A$, in which case no correction state is needed. The uniform certificate $\alpha_r \geq 1/d_A$ is unchanged. The proof of Theorem 4.7 uses the constant value $\lambda = 1/d_A$, which makes continuity of the construction immediate. Near points with $\mu(Z) = I_{A_0}/d_A$ the correction state $\nu(Z) = (I_{A_0}/d_A - \lambda(Z)\mu(Z))/(1 - \lambda(Z))$ becomes an indeterminate quotient: the weight $1 - \lambda(Z)$ tends to zero while the vanishing of the numerator depends on the approach direction, so a variable λ would require a separate gluing argument at exactly those points.

Corollary 4.9 (Antipodal cutoff at affine dimension)**.** *Assume $nd_B > 1$. Then $\alpha_r(\operatorname{Inst}_n(A, B)) = 0$ for all $r \geq D^{(n)}_{A,B}$.*

Proof. Choose affine coordinates $c : \mathfrak{A}^{(n)}_{A,B} \to \mathbb{R}^D$ (an affine isomorphism, hence injective). For any continuous $h : S^r \to \operatorname{Inst}_n(A, B)$ with $r \geq D$, restrict to an equatorial $S^D \subset S^r$; Borsuk–Ulam applied to $c \circ h : S^D \to \mathbb{R}^D$ gives u with $c(h(u)) = c(h(-u))$, hence $h(u) = h(-u)$ by injectivity. Thus $\alpha_D = 0$, and monotonicity (Proposition 4.3) extends this to all $r \geq D$. □

Theorem 4.10 (Observable antipodal sphere)**.** *Assume either (i) $n \geq 2$ and $d_A \geq 2$, or (ii) $n = 1$, $d_A \geq 2$, $d_B \geq 2$. Then there exists a continuous map $h : S^{d_A^2 - 2} \to \operatorname{Inst}_n(A, B)$ whose antipodal images have instrument-diamond distance 2; consequently*

$$\alpha_r(\operatorname{Inst}_n(A, B)) = 1 \qquad (0 \leq r \leq d_A^2 - 2).$$

For $n = 1$ this is the observable sphere of Ref. [2]. The construction is a continuous image of the sphere and may fail to be injective: H and its positive scalar multiples define the same instrument. Injectivity is not required by the width arguments.

Proof. First suppose $n \geq 2$. Fix a unit vector $|b_0\rangle \in B$ and let $\sigma = |b_0\rangle\langle b_0|$. For $H \in \mathrm{Herm}_0(A)$, $H \neq 0$, define

$$K(H) = \frac{H}{\|H\|_\infty}, \qquad E_\pm(H) = \frac{I_A \pm K(H)}{2}.$$

Then $E_\pm(H) \geq 0$ and $E_+(H) + E_-(H) = I_A$. Define an instrument by

$$\mathcal{I}_1^H(X) = \mathrm{Tr}(E_+(H)X)\sigma, \qquad \mathcal{I}_2^H(X) = \mathrm{Tr}(E_-(H)X)\sigma, \qquad \mathcal{I}_k^H(X) = 0 \ (k \geq 3);$$

the sum is the replacement channel $X \mapsto \mathrm{Tr}(X)\sigma$, so $\mathcal{I}^H \in \mathrm{Inst}_n(A, B)$. Since $\|K(H)\|_\infty = 1$, $K(H)$ has a unit eigenvector $|\psi\rangle$ with eigenvalue $\varepsilon \in \{+1, -1\}$. Then

$$\mathrm{Tr}(E_\varepsilon(H)|\psi\rangle\langle\psi|) = 1, \qquad \mathrm{Tr}(E_{-\varepsilon}(H)|\psi\rangle\langle\psi|) = 0,$$

where $E_{+1}(H) = E_+(H)$ and $E_{-1}(H) = E_-(H)$; since $E_\pm(-H) = E_\mp(H)$, the two effects are exchanged for $-H$. Hence the flagged outputs of $\mathcal{I}^H$ and $\mathcal{I}^{-H}$ on the input $|\psi\rangle\langle\psi|$ are $\sigma \otimes |1\rangle\langle 1|$ and $\sigma \otimes |2\rangle\langle 2|$ in one order or the other, which are orthogonal, so their trace norm difference is 2; by the diameter bound the instrument distance is exactly 2. The map $H \mapsto H/\|H\|_\infty$ is continuous on the Euclidean unit sphere of $\mathrm{Herm}_0(A) \cong \mathbb{R}^{d_A^2-1}$, so the construction gives a continuous image of $S^{d_A^2-2}$ with antipodal distance 2; equatorial restriction gives the result for all $0 \leq r \leq d_A^2 - 2$.

For $n = 1$ and $d_B \geq 2$, choose two orthogonal pure output states $\sigma_\pm = |\varphi_\pm\rangle\langle\varphi_\pm|$ and define the channel

$$\Phi^H(X) = \mathrm{Tr}(E_+(H)X)\sigma_+ + \mathrm{Tr}(E_-(H)X)\sigma_-.$$

The same eigenvector witness gives orthogonal outputs σ_+ and σ_-, hence distance 2, and the continuity argument is identical. ◻

Proposition 4.11 (Orthogonal join of flagged state spheres)**.** *Let $W_1, \ldots, W_\ell$ be pairwise orthogonal flag-block-diagonal subspaces of $B \otimes C_n$, and suppose that continuous maps $h_i : S^{r_i} \to \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ take values in states supported on W_i and satisfy*

$$\|h_i(u) - h_i(-u)\|_1 = 2 \qquad (u \in S^{r_i}).$$

Then there is a continuous map $H : S^{\sum_i r_i + \ell - 1} \to \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ with

$$\|H(y) - H(-y)\|_1 = 2$$

for every y. Composing with the replacement-instrument embedding gives the same antipodal separation in instrument diamond norm.

Proof. View the domain sphere as the unit sphere of $\bigoplus_{i=1}^{\ell} \mathbb{R}^{r_i+1}$. For a point $y = (y_1, \ldots, y_\ell)$ on it, write $\lambda_i = \|y_i\|_2^2$ and, on the blocks with $y_i \neq 0$, $u_i = y_i/\|y_i\|_2$; then define

$$H(y) = \sum_{i: y_i \neq 0} \lambda_i \, h_i(u_i),$$

reading the i-th summand as zero when $y_i = 0$. The weights satisfy $\sum_i \lambda_i = 1$, so $H(y)$ is a flagged state. Continuity at points where some $y_i = 0$ follows because the trace norm of the i-th summand is $\|\lambda_i h_i(u_i)\|_1 = \lambda_i \to 0$ as $y_i \to 0$, uniformly in the direction u_i. Because the supports W_i are pairwise orthogonal, the difference $H(y) - H(-y)$ is block diagonal with respect to $\bigoplus_i W_i$, so

$$\|H(y) - H(-y)\|_1 = \sum_i \lambda_i \|h_i(u_i) - h_i(-u_i)\|_1 = 2\sum_i \lambda_i = 2.$$

Lemma 4.4 transfers the equality to replacement instruments. ◻

Proposition 4.12 (Instrument-level join with replacement spheres)**.** *Let $W \subseteq B \otimes C_n$ be a flag-block-diagonal subspace. Let $h : S^{r_1} \to \mathrm{Inst}_n(A, B)$ be continuous and such that*

1. *for every input state ρ on $A_0 \otimes A$ and every $u \in S^{r_1}$, the flagged output $(\mathrm{id}_{A_0} \otimes \widehat{h(u)})(\rho)$ is supported on $A_0 \otimes W$, and*
2. *$\|h(u) - h(-u)\|_{\diamond,\mathrm{inst}} = 2$ for every $u \in S^{r_1}$.*

Let $\tau : S^{r_2} \to \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ be continuous, take values in states supported on the per-block orthogonal complement $W^{\perp}$, and satisfy $\|\tau(v) - \tau(-v)\|_1 = 2$ for every $v \in S^{r_2}$. Then, on the unit sphere of $\mathbb{R}^{r_1+1} \oplus \mathbb{R}^{r_2+1}$,

$$\mathcal{I}(y) = \|y_1\|_2^2\, h\!\left(\frac{y_1}{\|y_1\|_2}\right) + \|y_2\|_2^2\, \vec{\mathcal{R}}_{\tau(y_2/\|y_2\|_2)}$$

(with a summand read as zero when the corresponding coordinate block vanishes) defines a continuous map $S^{r_1+r_2+1} \to \mathrm{Inst}_n(A, B)$ with

$$\|\mathcal{I}(y) - \mathcal{I}(-y)\|_{\diamond,\mathrm{inst}} = 2 \qquad (y \in S^{r_1+r_2+1}).$$

Proof. Both summands are convex combinations of instruments, hence instruments, so $\mathcal{I}(y) \in \mathrm{Inst}_n(A, B)$; the instrument body is bounded (every flagged channel has diamond norm 1), so each summand tends to zero in norm as its weight tends to zero, uniformly over the sphere, which gives continuity on the coordinate spheres $\{y_i = 0\}$ exactly as in Proposition 4.11. Fix y and write $\lambda = \|y_1\|_2^2$. The endpoint cases reduce to single components: at $\lambda = 0$ the pair is the replacement pair, at distance 2 by Lemma 4.4, and at $\lambda = 1$ it is the observable pair, at distance 2 by hypothesis (ii). Assume $0 < \lambda < 1$, write $u = y_1/\|y_1\|_2$ and $v = y_2/\|y_2\|_2$, and set $\vec{\Delta}_1 = h(u) - h(-u)$ and $\vec{\Delta}_2 = \vec{\mathcal{R}}_{\tau(v)} - \vec{\mathcal{R}}_{\tau(-v)}$, so that $\mathcal{I}(y) - \mathcal{I}(-y) = \lambda\vec{\Delta}_1 + (1-\lambda)\vec{\Delta}_2$. The set of input states is compact and $\rho \mapsto \|(\mathrm{id} \otimes \vec{\Delta}_1)(\rho)\|_1$ is continuous with supremum $\|\vec{\Delta}_1\|_{\diamond,\mathrm{inst}} = 2$, so there is an input state ρ_* with $\|(\mathrm{id} \otimes \vec{\Delta}_1)(\rho_*)\|_1 = 2$. For this same input, Lemma 4.4 gives $(\mathrm{id} \otimes \vec{\Delta}_2)(\rho_*) = \mathrm{Tr}_A(\rho_*) \otimes (\tau(v) - \tau(-v))$, of trace norm $\|\tau(v) - \tau(-v)\|_1 = 2$, independently of ρ_*. By (i) applied to $h(u)$ and $h(-u)$, the Hermitian operator $(\mathrm{id} \otimes \vec{\Delta}_1)(\rho_*)$ is supported on $A_0 \otimes W$, while $(\mathrm{id} \otimes \vec{\Delta}_2)(\rho_*)$ is supported on $A_0 \otimes W^{\perp}$; orthogonal supports make the trace norm additive, so

$$\big\|(\mathrm{id} \otimes (\mathcal{I}(y) - \mathcal{I}(-y)))(\rho_*)\big\|_1 = \lambda \cdot 2 + (1-\lambda) \cdot 2 = 2.$$

Hence $\|\mathcal{I}(y) - \mathcal{I}(-y)\|_{\diamond,\mathrm{inst}} \geq 2$, and the reverse inequality is the diameter bound (Proposition 3.9). ☐

Remark 4.13 (Why the join carries at most one input-dependent sphere)**.** *Proposition 4.12 joins one* input-dependent *sphere with replacement spheres: the replacement components attain their maximal separation* 2 *on* every *input, so an input that exposes the input-dependent component simultaneously exposes all of them. Two input-dependent components could not be joined by this mechanism, since their separating inputs (eigenvectors of the respective observables) vary with the sphere parameter and need not coincide; this is why exactly one observable sphere enters Corollary 4.14 below.*

Corollary 4.14 (Observable–replacement join dimension)**.** *Under the hypotheses of Theorem 4.10, the flagged outputs of the observable sphere are supported on the flagged subspace*

$$W_{\mathrm{obs}} = \mathrm{span}\{|b_0\rangle \otimes |1\rangle,\ |b_0\rangle \otimes |2\rangle\}$$

in the case $n \geq 2$ (with the pure state $\sigma = |b_0\rangle\langle b_0|$), and on the two-dimensional flagged subspace $W_{\mathrm{obs}} = \mathrm{span}\{|\varphi_+\rangle, |\varphi_-\rangle\}$ in the case $n = 1$, where $\sigma_{\pm} = |\varphi_{\pm}\rangle\langle\varphi_{\pm}|$ are the two orthogonal output states of the channel construction. Let $W_{\mathrm{obs}}^{\perp}$ denote the per-block orthogonal complement of W_{obs}:

when $n \geq 2$ its first two blocks are $\{|b_0\rangle\}^\perp \subseteq B$ and its remaining $n-2$ blocks are all of B; when $n = 1$ it is the orthogonal complement of W_{obs} in B. The real vector space of flag-block-diagonal Hermitian operators supported on $W_{\mathrm{obs}}^\perp$ has dimension

$$N_{\mathrm{join}} = \begin{cases} 2(d_B-1)^2 + (n-2)d_B^2, & n \geq 2, \\ (d_B-2)^2, & n = 1. \end{cases}$$

If $N_{\mathrm{join}} \geq 1$, then

$$\alpha_r(\mathrm{Inst}_n(A,B)) = 1 \qquad (0 \leq r \leq d_A^2 + N_{\mathrm{join}} - 3).$$

The boundary value $N_{\mathrm{join}} = 1$, which occurs only for $n = 1$, $d_B = 3$, is degenerate: the replacement sphere $S^{N_{\mathrm{join}}-2} = S^{-1}$ is empty, the join reduces to the observable sphere alone, and the displayed formula reproduces its index $d_A^2 - 2$. For $n = 1$ this is the observable–replacement join of Ref. [2].

Proof. The observable sphere $h : S^{d_A^2-2} \to \mathrm{Inst}_n(A,B)$ of Theorem 4.10 is continuous and has antipodal instrument-diamond distance 2; on every input ρ, each outcome component of its flagged output is a positive operator on A_0 tensored with $\sigma \otimes |k\rangle\langle k|$, $k \in \{1,2\}$ (respectively with σ_+ or σ_- when $n = 1$), and the remaining outcomes vanish, so all flagged outputs are supported on $A_0 \otimes W_{\mathrm{obs}}$, i.e. hypothesis (i) of Proposition 4.12 holds with $W = W_{\mathrm{obs}}$. The per-block complement $W_{\mathrm{obs}}^\perp$ is itself flag-block-diagonal, and the Jordan construction of Theorem 4.5, applied to the traceless part of the real vector space of flag-block-diagonal Hermitian operators supported on $W_{\mathrm{obs}}^\perp$, produces a continuous sphere $\tau : S^{N_{\mathrm{join}}-2} \to \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ of states supported on $W_{\mathrm{obs}}^\perp$ with $\|\tau(v) - \tau(-v)\|_1 = 2$; the condition $N_{\mathrm{join}} \geq 2$ guarantees nonemptiness of this sphere, while at $N_{\mathrm{join}} = 1$ the traceless block space is trivial, the sphere is empty, and the observable sphere alone realises the displayed range, so the statement holds for $N_{\mathrm{join}} \geq 1$. Proposition 4.12 joins the two spheres into a continuous map

$$S^{(d_A^2-2)+(N_{\mathrm{join}}-2)+1} = S^{d_A^2+N_{\mathrm{join}}-3} \longrightarrow \mathrm{Inst}_n(A,B)$$

with antipodal instrument-diamond distance 2; equatorial restriction supplies every smaller index, and $\alpha_r \leq 1$ always by the diameter bound (Proposition 3.9). □

Remark 4.15 (Structured partitions)**.** *The ℓ-fold join of Proposition 4.11 may also be applied to arbitrary pairwise orthogonal output–flag blocks (B_i, n_i) with $\sum_i b_i \leq d_B$, $\sum_i n_i = n$, producing a sphere of dimension $\sum_i (n_i b_i^2 - 1) - 1$ whenever $n_i b_i^2 \geq 2$ for every retained block. Taking one block with $b_1 = d_B$, $n_1 = n$ recovers the leading dimension $nd_B^2 - 2$; no partition improves Theorem 4.5.*

Definition 4.16 (Certified one-sphere dimension)**.** *Let*

$$L_{\mathrm{rep}} = nd_B^2 - 2,$$

valid whenever $nd_B > 1$. If either $n \geq 2, d_A \geq 2$, or $n = 1, d_A \geq 2, d_B \geq 2$, set $L_{\mathrm{obs}} = d_A^2 - 2$; otherwise set $L_{\mathrm{obs}} = -1$. If the observable sphere is valid and $N_{\mathrm{join}} \geq 1$ (Corollary 4.14), set $L_{\mathrm{join}} = d_A^2 + N_{\mathrm{join}} - 3$ (at the degenerate value $N_{\mathrm{join}} = 1$, occurring only for $n = 1$, $d_B = 3$, this equals L_{obs}); otherwise $L_{\mathrm{join}} = -1$. Finally define

$$L_{A,B}^{(n)} = \min\left\{ D_{A,B}^{(n)} - 1,\ \max\{L_{\mathrm{rep}}, L_{\mathrm{obs}}, L_{\mathrm{join}}\} \right\},$$

where invalid terms (-1) are ignored. If $d_A = 1$, the observable and join terms are invalid and $L_{1,B}^{(n)} = nd_B^2 - 2 = D_{1,B}^{(n)} - 1$.

Definition 4.17 (Certified lower envelope)**.** *Assume $nd_B > 1$. Let*

$$\rho_n = \frac{2}{nd_B \min\{d_A, d_B\}}, \qquad \gamma_{A,B}^{(n)} = \max\Big\{\rho_n, \frac{1}{d_A}\Big\}.$$

Define

$$\beta_r^{\mathrm{best},(n)}(A,B) = \begin{cases} R_n = 2\big(1 - \frac{1}{nd_B s}\big), & r = 0, \\ 1, & 1 \le r \le L_{A,B}^{(n)}, \\ \gamma_{A,B}^{(n)}, & L_{A,B}^{(n)} < r < D_{A,B}^{(n)}, \\ 0, & r \ge D_{A,B}^{(n)}. \end{cases}$$

When $L_{A,B}^{(n)} = D_{A,B}^{(n)} - 1$ the middle range is empty (this is always the case for $d_A = 1$). The value at $r = 0$ is the exact constant-code width of Corollary 5.16: the envelope is upgraded there from the one-sphere value 1 to R_n, so the lower and upper envelopes coincide at $r = 0$. Since $nd_B s \ge 2$ gives $R_n \ge 1$ and $d_A \ge 1$ gives $0 < \gamma_{A,B}^{(n)} \le 1$, the displayed envelope is non-increasing in r. Every component is rigorously certified by Theorems 4.5, 4.7, 4.10, Corollary 4.14, and Theorem 3.3, and, at $r = 0$, Corollary 5.16.

Remark 4.18 (Resource interpretation of the two breakpoints)**.** *The two breakpoints of the envelope have a direct resource interpretation. A flagged output state has $nd_B^2 - 1$ affine parameters, while an input-dependent instrument has $d_A^2(nd_B^2 - 1)$. The first regime ($r \le L_{A,B}^{(n)}$) is already obstructed by preparing and recording one classical–quantum output state — augmented by the observable constructions, which obstruct input-dependent effect readout even when the output state is fixed. The second regime concerns the additional dependence of all outcome branches on the input operator. Increasing the number of outcomes therefore has two opposite effects: it linearly enlarges the global representation space, while dividing the local positivity margin of the uniform instrument by n.*

Remark 4.19 (Two mechanisms behind the intermediate floor)**.** *The intermediate floor $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$ combines two certificates of different geometric origin, and the maximum is the correct combination because each term is individually a valid lower bound on every width in the intermediate regime. The term $\rho_n = 2/(nd_B s)$ is a local statement: a ball of radius ρ_n about the uniform instrument exists, and the antipodal pairs inside a centrally symmetric ball certify error at least ρ_n via the radial map (Corollary 5.5); it decays with n because the uniform trace budget is split among n outcomes. The term $1/d_A$ is a global statement: a sphere spanning all $D_{A,B}^{(n)}$ directions of the body exists whose antipodes are separated by $2/d_A$ (Theorem 4.7); the value $1/d_A$ is the largest mixing weight at which two orthogonal flagged Choi states with a common input marginal can be completed to instruments uniformly over all directions, the direction-dependent optimum $1/(d_A\|\mu(Z)\|_\infty)$ of Remark 4.8 attaining its worst-case value $1/d_A$ exactly when the common marginal is pure. The two terms trade off at $n \approx 2d_A/(d_B \min\{d_A, d_B\})$: below this threshold the ball term dominates the maximum, above it the input-dimension term $1/d_A$ is the floor; in particular, whenever $d_A \le d_B$ the floor is $1/d_A$ for every $n \ge 2$. This is the geometric reason the intermediate obstruction carries an n-independent component. The $n = 1$ value $\gamma_{A,B}^{(1)} = \max\{2/(d_B \min\{d_A, d_B\}), 1/d_A\}$ is the intermediate envelope value of Ref. [2].*

5 Compression widths: upper bounds and exact thresholds

The spheres of the preceding section certify that certain widths are large; this section supplies the constructive upper bounds and the exact threshold where the width collapses to zero. The two sides bracket every intermediate width in an explicit interval. The widths defined below are continuous-encoding analogues of the classical Kolmogorov widths of a compact set in a normed space [8], with the approximating linear subspaces of the classical theory replaced by arbitrary continuous encodings into $\mathbb{R}^r$.

Convention 5.1 (Continuous encoding, unrestricted decoding)**.** *Throughout the compression-width results, an encoder is required to be continuous while the decoder is unrestricted unless explicitly stated otherwise. This convention is essential for the Helly-type upper bounds; where a continuous decoder is available we state this separately.*

Definition 5.2 (Euclidean instrument width)**.**

$$\delta^{\diamond}_{r,\mathrm{inst}}(A,B) = \inf_{\substack{f:\mathrm{Inst}_n(A,B)\to\mathbb{R}^r \ \textit{continuous} \\ g:\mathbb{R}^r\to\mathrm{Inst}_n(A,B)}} \sup_{\mathcal{I}\in\mathrm{Inst}_n(A,B)} \|g(f(\mathcal{I}))-\mathcal{I}\|_{\diamond,\mathrm{inst}}.$$

For compact metric $K\subseteq L$ we also use $\delta^{\mathrm{c}}_r(K;L)$, the analogous width with continuous encoders into $\mathbb{R}^r$ and arbitrary (not necessarily continuous) decoders into L.

Remark 5.3 (Decoder codomain does not weaken the lower bounds)**.** *Since* $\mathrm{RepInst}_n(A,B)\subset\mathrm{Inst}_n(A,B)\subset\mathfrak{A}^{(n)}_{A,B}$*, and the triangle inequality* $\|h(u)-\Psi\|_{\diamond,\mathrm{inst}}+\|h(-u)-\Psi\|_{\diamond,\mathrm{inst}}\geq\|h(u)-h(-u)\|_{\diamond,\mathrm{inst}}$ *holds for every* Ψ *in the ambient vector space* $\bigoplus_k \mathrm{L}(A,B)$*, the lower bounds of Corollaries 5.5 and 5.6 (stated below) are proved for decoders with values in the full affine space* $\mathfrak{A}^{(n)}_{A,B}$ *— a strictly stronger statement than for instrument-valued decoders. Hence*

$$\begin{aligned}\beta^{\mathrm{best},(n)}_r(A,B) &\leq \delta^{\mathrm{c}}_r\big(\mathrm{Inst}_n(A,B);\mathfrak{A}^{(n)}_{A,B}\big)\\ &\leq \delta^{\mathrm{c}}_r\big(\mathrm{Inst}_n(A,B);\mathrm{Inst}_n(A,B)\big)=\delta^{\diamond}_{r,\mathrm{inst}}(A,B)\leq U^{(n)}_r(A,B).\end{aligned}$$

Allowing unphysical affine decoders cannot circumvent the certified lower envelope; and because the optimal constant-code centre $\mathcal{I}_*$ *is itself an instrument (Theorem 3.8), the constant-code width coincides over both codomains (Corollary 5.16), so every upper bound of this paper is admissible with a physical instrument decoder. The same codomain convention underlies the Hermiticity-preserving trace-preserving decoder bounds of Ref. [2].*

Corollary 5.4 (Certified width bracket)**.** *Assume* $nd_B>1$*. For* $0\leq r<D^{(n)}_{A,B}$*,*

$$\beta^{\mathrm{best},(n)}_r(A,B)\leq\delta^{\diamond}_{r,\mathrm{inst}}(A,B)\leq U^{(n)}_r(A,B),$$

where $U^{(n)}_r$ *is defined in Definition 5.14. For* $r\geq D^{(n)}_{A,B}$ *the width is zero.*

Proof. At $r=0$ every encoder is constant, and Corollary 5.16 gives $\delta^{\diamond}_{0,\mathrm{inst}}(A,B)=R_n=\beta^{\mathrm{best},(n)}_0(A,B)=U^{(n)}_0(A,B)$, so the bracket is exact at $r=0$. For $1\leq r\leq L^{(n)}_{A,B}$ the replacement sphere, observable sphere, or observable–replacement join supplies a continuous map $h:S^r\to\mathrm{Inst}_n(A,B)$ with antipodal distance 2; composing with any continuous encoder f gives $f\circ h: S^r\to\mathbb{R}^r$, and Borsuk–Ulam produces u with $f(h(u))=f(h(-u))$. The decoder returns one reconstruction for the pair, and the triangle inequality forces error at least $1=\beta^{\mathrm{best},(n)}_r$. The argument applies verbatim to decoders valued in the ambient affine space $\mathfrak{A}^{(n)}_{A,B}$ (Remark 5.3), which only strengthens the lower bound. For $L^{(n)}_{A,B}<r<D^{(n)}_{A,B}$, two independent certificates hold: the centred ball gives antipodal separation $2\rho_n$ (via the radial map of Corollary 5.5, which needs only $r+1\leq D_n$), and Theorem 4.7 gives separation $2/d_A$. Choose the radial map if $\rho_n\geq 1/d_A$ and the full-direction map if $1/d_A\geq\rho_n$; the chosen map has separation $2\gamma^{(n)}_{A,B}$, hence error at least $\gamma^{(n)}_{A,B}$. The upper terms are Theorem 3.8, Theorem 5.11 and Proposition 5.8 (stated below); exact reconstruction for $r\geq D^{(n)}_{A,B}$ is Theorem 5.7. □

For a Hausdorff latent space Y, recall the deleted product

$$F_2(Y) = \{(y_0, y_1) \in Y^2 : y_0 \neq y_1\}$$

with the involution exchanging the two entries, and let

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) = \sup\bigl\{k \geq 0 : \exists \text{ equivariant } S^k \to F_2(Y)\bigr\},$$

with value -1 when $F_2(Y)$ is empty and possible value $+\infty$. For a metric space Y, $F_2(Y)$ carries the subspace topology of Y^2 and the free $\mathbb{Z}_2$-action; Hausdorffness is not needed for the deleted product itself, but is natural to make the diagonal closed. If $\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) = +\infty$, no integer q satisfies $q > \operatorname{coind}_{\mathbb{Z}_2} F_2(Y)$ and the corollary below is vacuous. For $r \geq 1$, if Y admits a continuous injection $\jmath : Y \to \mathbb{R}^r$, the normalised difference map

$$(y_0, y_1) \longmapsto \frac{\jmath(y_0) - \jmath(y_1)}{\|\jmath(y_0) - \jmath(y_1)\|_2}$$

is continuous and equivariant $F_2(Y) \to S^{r-1}$, so by Borsuk–Ulam [15], $\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) \leq r - 1$.

Corollary 5.5 (Intrinsic latent-space instrument hierarchy)**.** *Assume $nd_B > 1$. Let Y be Hausdorff and let $q \in \mathbb{N}_0$ satisfy*

$$q > \operatorname{coind}_{\mathbb{Z}_2} F_2(Y).$$

Then every continuous encoder $\mathrm{Enc} : \mathrm{Inst}_n(A, B) \to Y$ *and every decoder* $\mathrm{Dec} : Y \to \mathfrak{A}^{(n)}_{A,B}$ *satisfy*

$$\sup_{\mathcal{I} \in \mathrm{Inst}_n(A,B)} \|\operatorname{Dec}(\operatorname{Enc}(\mathcal{I})) - \mathcal{I}\|_{\diamond,\mathrm{inst}} \geq \beta_q^{\mathrm{best},(n)}(A, B).$$

Proof. If $q \geq D^{(n)}_{A,B}$, then $\beta_q^{\mathrm{best},(n)}(A, B) = 0$ and the claimed inequality is trivial; assume henceforth $q < D^{(n)}_{A,B}$. If $q = 0$, the hypothesis $q > \operatorname{coind}_{\mathbb{Z}_2} F_2(Y)$ forces $F_2(Y) = \varnothing$, i.e. Y is a singleton; the decoder returns a single point $\Psi \in \mathfrak{A}^{(n)}_{A,B}$, and the centre-optimality clause of Theorem 3.8 gives $\sup_{\mathcal{I}} \|\Psi - \mathcal{I}\|_{\diamond,\mathrm{inst}} = R_{\mathrm{inst}}(\Psi) \geq R_n = \beta_0^{\mathrm{best},(n)}(A, B)$. If $1 \leq q \leq L^{(n)}_{A,B}$, restrict the perfectly separated sphere supplied by Theorem 4.5, Theorem 4.10, or Corollary 4.14 to an equatorial S^q. If $L^{(n)}_{A,B} < q < D^{(n)}_{A,B}$, then $q + 1 \leq D^{(n)}_{A,B}$; choose a $(q + 1)$-dimensional subspace $W \subseteq \mathfrak{V}^{(n)}_{A,B}$ and use the radial map

$$h(u) = \mathcal{I}_* + \rho_n \frac{u}{\|u\|_{\diamond,\mathrm{inst}}}$$

on an auxiliary Euclidean sphere $S^q \subset W$ (the denominator $\|u\|_{\diamond,\mathrm{inst}}$ is a continuous and strictly positive function on the compact Euclidean sphere, so the normalisation is continuous). Theorem 3.3 makes the radial map instrument-valued with antipodal separation $2\rho_n$, and Theorem 4.7 independently supplies a map with separation $2/d_A$; choose the radial map when $\rho_n \geq 1/d_A$ and the full-direction map otherwise, so that the chosen map has antipodal separation $2\gamma^{(n)}_{A,B}$. In both regimes, if the two latent codes $\mathrm{Enc}(h(u))$ and $\mathrm{Enc}(h(-u))$ were distinct for every u, then $u \mapsto (\mathrm{Enc}(h(u)), \mathrm{Enc}(h(-u)))$ would be an equivariant map $S^q \to F_2(Y)$, contradicting the coindex hypothesis. An antipodal pair therefore has one code; the decoder returns one reconstruction for it, and the triangle inequality gives half the separation, namely $\beta_q^{\mathrm{best},(n)}(A, B)$. $\square$

Corollary 5.6 (Continuous instrument-compression hierarchy)**.** *Assume $nd_B > 1$, and let Y admit a continuous injection into $\mathbb{R}^r$. Then, for every continuous encoder* $\mathrm{Enc} : \mathrm{Inst}_n(A, B) \to Y$ *and every decoder* $\mathrm{Dec} : Y \to \mathfrak{A}^{(n)}_{A,B}$,

$$\sup_{\mathcal{I} \in \mathrm{Inst}_n(A,B)} \|\operatorname{Dec}(\operatorname{Enc}(\mathcal{I})) - \mathcal{I}\|_{\diamond,\mathrm{inst}} \geq \beta_r^{\mathrm{best},(n)}(A, B).$$

Proof. If $r = 0$ then Y has at most one point and $F_2(Y) = \emptyset$ has coindex -1; if $r \geq 1$ the normalised difference map gives $\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) \leq r - 1$ as above. Apply Corollary 5.5 with $q = r$. □

Theorem 5.7 (Zero-error threshold, with continuous decoding at the threshold)**.** *For Euclidean latent spaces and under Convention 5.1,*

$$\delta^{\diamond}_{r,\mathrm{inst}}(A, B) = 0 \quad \iff \quad r \geq D^{(n)}_{A,B}.$$

Moreover, for every $r \geq D^{(n)}_{A,B}$, exact decoding may be chosen continuous.

Proof. For $r \geq D^{(n)}_{A,B}$, choose affine coordinates $c : \mathfrak{A}^{(n)}_{A,B} \to \mathbb{R}^D$ (an affine isomorphism) and restrict to the compact convex body $K = \mathrm{Inst}_n(A, B)$; its image $C = c(K) \subset \mathbb{R}^D$ is compact and convex. The encoder is $f = c|_K$, padded with zeros if $r > D$. Let $\Pi_C : \mathbb{R}^D \to C$ be the Euclidean nearest-point projection: for a nonempty closed convex subset of Euclidean space the metric projection is single-valued and 1-Lipschitz, hence continuous. Define the decoder $g(z) = c^{-1}(\Pi_C(z_1, \ldots, z_D))$, ignoring any padding. Then $g(f(x)) = x$ for every $x \in K$ and g is continuous. For the converse, if $r < D^{(n)}_{A,B}$ then $\beta^{\mathrm{best},(n)}_r(A, B) > 0$ (Corollary 5.6), so exact reconstruction is impossible even with an arbitrary decoder; the collision $f(h(u)) = f(h(-u))$ forces one decoder value to cover two distinct instruments, using only that the *code values coincide*, with no regularity assumption on the decoder. □

Proposition 5.8 (Convex-projection instrument bound)**.** *For $0 \leq r \leq D^{(n)}_{A,B}$,*

$$\delta^{\diamond}_{r,\mathrm{inst}}(A, B) \leq \frac{2(D^{(n)}_{A,B} - r)}{D^{(n)}_{A,B} - r + 1}.$$

Proof. The argument is the convex-projection estimate of Ref. [2], specialised to the present body and included here for completeness. Write $K = \mathrm{Inst}_n(A, B)$ and $D = D^{(n)}_{A,B}$; Proposition 3.9 gives $\operatorname{diam} K = 2$. If $r = D$, any affine coordinate system on $\operatorname{aff} K$, padded by zeros and inverted on its compact image (with an arbitrary extension off the image), reconstructs K exactly. Hence assume $0 \leq r < D$ and set $k = D - r$. Fix affine coordinates on $\operatorname{aff} K$ and let $f : K \to \mathbb{R}^r$ retain the first r of them. Every nonempty fibre $F_y = f^{-1}(y)$ is compact and convex, has diameter $\Delta_y \leq 2$, and has affine dimension at most k.

Set $R_y = k\Delta_y/(k+1)$ and work inside the normed affine space $\operatorname{aff}(F_y)$. Every $k+1$ members of the compact family $\{F_y\} \cup \{F_y \cap \overline{B}(x, R_y) : x \in F_y\}$ intersect: given $m \leq k + 1$ centres $x_1, \ldots, x_m \in F_y$, their mean c lies in F_y by convexity and satisfies $\|c - x_i\| \leq \frac{m-1}{m}\Delta_y \leq R_y$ by the triangle inequality, while a subfamily containing F_y leaves at most k balls, for which the same estimate applies. By Helly's theorem [16] and the finite-intersection property, some point $c_y \in F_y$ lies within distance R_y of every point of F_y. The decoder g, equal to c_y on $f(K)$ and to any fixed instrument elsewhere, requires no regularity (Definition 5.2) and achieves

$$\|g(f(x)) - x\| \leq \frac{k}{k+1}\Delta_{f(x)} \leq \frac{2(D^{(n)}_{A,B} - r)}{D^{(n)}_{A,B} - r + 1}$$

for every $x \in K$, which proves the claim. □

Remark 5.9 (Nonconstructive Helly decoder)**.** *The decoder in Proposition 5.8 is selected via Helly's theorem and compactness for each fibre; it is guaranteed to take values in the fibre and to achieve error $\leq 2k/(k+1)$; the decoder need not be continuous, and Definition 5.2 imposes no continuity requirement on decoders. Throughout, "certified" refers to proved upper bounds; it does not imply that the decoder is continuous or algorithmic. If both encoder and decoder* *were required to be continuous, the same zero-error threshold $r \geq D^{(n)}_{A,B}$ remains valid: the upper direction is Theorem 5.7 (which supplies a continuous decoder), and the Borsuk–Ulam lower bound applies a fortiori to continuous decoders. The subcritical upper bounds, however, become existence statements whose Helly-selected decoders may fail continuity. A quantitative error bound for Lipschitz decoders, in terms of the antipodal profile of the reachable latent image, is developed in Ref. [2].*

Lemma 5.10 (Exact pinching norm)**.** *For an orthogonal decomposition $H = \bigoplus_{j=1}^{\ell} H_j$ into nonzero blocks and the channel $\mathcal{P}_H(X) = \sum_j P_j X P_j$,*

$$\| \mathrm{id}_H - \mathcal{P}_H \|_\diamond = 2\Big(1 - \frac{1}{\ell}\Big).$$

Proof. The identity is the channel-level pinching bound of Ref. [2]; the short argument is reproduced here for completeness. Let $\omega = e^{2\pi i/\ell}$ and $U_m = \sum_{j=1}^{\ell} \omega^{m(j-1)} P_j$. Averaging the conjugation channels $\mathcal{U}_{U_m}(X) = U_m X U_m^\dagger$ over m and applying $\frac{1}{\ell}\sum_{m=0}^{\ell-1} \omega^{m(i-j)} = \delta_{ij}$ recovers $\mathcal{P}_H$ exactly. The triangle inequality together with $\| \mathrm{id}_H - \mathcal{U}_{U_m}\|_\diamond \leq 2$ then yields

$$\| \mathrm{id}_H - \mathcal{P}_H \|_\diamond \leq \frac{1}{\ell}\sum_{m=1}^{\ell-1} \| \mathrm{id}_H - \mathcal{U}_{U_m}\|_\diamond \leq 2\Big(1 - \frac{1}{\ell}\Big).$$

For the reverse bound, pick unit vectors $|e_j\rangle \in H_j$ and form the uniform superposition $|e\rangle = \ell^{-1/2}\sum_j |e_j\rangle$. A direct computation gives $(\mathrm{id}_H - \mathcal{P}_H)(|e\rangle\langle e|) = \frac{1}{\ell}(J_\ell - I_\ell)$ on the span of the $|e_j\rangle$, where J_ℓ denotes the all-ones $\ell \times \ell$ matrix. The spectrum of J_ℓ is $\{\ell, 0, \ldots, 0\}$, so $\frac{1}{\ell}(J_\ell - I_\ell)$ has spectrum $\{(\ell-1)/\ell, -1/\ell, \ldots, -1/\ell\}$ and trace norm $2(1 - 1/\ell)$, which is attained by this input. ☐

Theorem 5.11 (Input- and output-pinching for instruments)**.** *The following bounds hold.*

1. *If $B = \bigoplus_{j=1}^{\ell_{\mathrm{out}}} B_j$, $b_j = \dim B_j > 0$, and*

$$r^{(n)}_{\mathrm{out}} = d_A^2\Big(n\sum_j b_j^2 - 1\Big),$$

then, for $r \geq r^{(n)}_{\mathrm{out}}$,

$$\delta^\diamond_{r,\mathrm{inst}}(A, B) \leq 2\Big(1 - \frac{1}{\ell_{\mathrm{out}}}\Big).$$

2. *If $A = \bigoplus_{i=1}^{k_{\mathrm{in}}} A_i$, $a_i = \dim A_i > 0$, and*

$$r^{(n)}_{\mathrm{in}} = \Big(\sum_i a_i^2\Big)(n d_B^2 - 1),$$

then, for $r \geq r^{(n)}_{\mathrm{in}}$,

$$\delta^\diamond_{r,\mathrm{inst}}(A, B) \leq 2\Big(1 - \frac{1}{k_{\mathrm{in}}}\Big).$$

Proof. For output pinching, the flagged Choi operators have one Hermitian block on $A_0 \otimes B_j$ for every outcome and output block, of ambient real dimension $n d_A^2 \sum_j b_j^2$. The summed marginal map onto $\mathrm{Herm}(A_0)$ is surjective: for prescribed $H \in \mathrm{Herm}(A_0)$ place $H \otimes I_{B_1}/b_1$ in one outcome block and set the others to zero, since $\mathrm{Tr}_{B_1}(H \otimes I_{B_1}/b_1) = H$. Thus the marginal rank is d_A^2, and the uniform depolarising instrument is invariant under output pinching and positive definite on every retained block, hence a relative interior point of the pinned body; so the output-pinched body has affine dimension $r_{\text{out}}^{(n)}$. Encode the output-pinched instrument by affine coordinates and invert on the image. The reconstructed instrument is defined componentwise by

$$\widetilde{\mathcal{I}}_k = \mathcal{P}_B \circ \mathcal{I}_k \qquad (k = 1, \ldots, n),$$

equivalently, its flagged channel is $(\mathcal{P}_B \otimes \mathrm{id}_{C_n}) \circ \widehat{\mathcal{I}}$; by composition submultiplicativity of the diamond norm [12] and Lemma 5.10,

$$\|\mathcal{I} - \widetilde{\mathcal{I}}\|_{\diamond,\text{inst}} = \|\widehat{\mathcal{I}} - (\mathcal{P}_B \otimes \mathrm{id}_{C_n}) \circ \widehat{\mathcal{I}}\|_\diamond \le \|\,\mathrm{id}_B - \mathcal{P}_B\|_\diamond \cdot \|\widehat{\mathcal{I}}\|_\diamond = 2\Big(1 - \frac{1}{\ell_{\text{out}}}\Big).$$

For input pinching, use the basis-conjugate decomposition $A_0 = \bigoplus_i \overline{A}_i$, where $\overline{A}_i$ is the conjugate subspace in A_0 of the block $A_i \subseteq A$, and let

$$\mathcal{P}_{\overline{A}_0}(Y) = \sum_i P_{\overline{A}_i} Y P_{\overline{A}_i}$$

be the block pinching on A_0. With the fixed Choi basis, the input pinching channel $\mathcal{P}_A = \sum_i P_i(\cdot)P_i$ induces on Choi operators

$$\widehat{J}_{\Phi\circ\mathcal{P}_A} = (\mathcal{P}_{\overline{A}_0} \otimes \mathrm{id}_B)\widehat{J}_\Phi.$$

The block-diagonal flagged Choi space has dimension $nd_B^2 \sum_i a_i^2$, and the blockwise marginal map is surjective: for prescribed $H_i \in \mathrm{Herm}(\overline{A}_i)$ place $H_i \otimes I_B/d_B$ in one outcome block. Thus the constraints have total rank $\sum_i a_i^2$; the uniform depolarising instrument is invariant under input pinching (I_{A_0} is block diagonal with respect to the conjugate decomposition) and remains a positive-definite relative interior point, so the affine dimension is $r_{\text{in}}^{(n)}$. Encoding $\mathcal{I} \circ \mathcal{P}_A$ by affine coordinates and inverting on the image gives

$$\|\mathcal{I} - \mathcal{I} \circ \mathcal{P}_A\|_{\diamond,\text{inst}} \le \|\widehat{\mathcal{I}}\|_\diamond \cdot \|\,\mathrm{id}_A - \mathcal{P}_A\|_\diamond = 2\Big(1 - \frac{1}{k_{\text{in}}}\Big).$$

Padding by zero coordinates handles larger r in both cases. ☐

Remark 5.12 (Pinching bounds are certified, not exact)**.** *The bounds of Theorem 5.11 are certified upper bounds on the compression width: they are achieved by the explicit pinching constructions above; the equalities $\delta^\diamond_{r,\text{inst}}(A,B) = 2(1 - 1/\ell_{\text{out}})$ and $\delta^\diamond_{r,\text{inst}}(A,B) = 2(1 - 1/k_{\text{in}})$ are not established at any r. For $n = 1$ the thresholds reduce to $r_{\text{out}}^{(1)} = d_A^2(\sum_j b_j^2 - 1)$ and $r_{\text{in}}^{(1)} = (\sum_i a_i^2)(d_B^2 - 1)$, the balanced-pinching thresholds of Ref. [2].*

Lemma 5.13 (Balanced partitions minimise squared block sums)**.** *For $1 \le \ell \le d$ write $d = m\ell + t$, $0 \le t < \ell$. Among all partitions of d into ℓ positive parts, the sum of squared parts is minimised by the balanced partition with $\ell - t$ parts equal to m and t parts equal to $m + 1$, and the minimum is*

$$Q_\ell(d) = (\ell - t)m^2 + t(m+1)^2.$$

Proof. If two block sizes satisfy $a \ge b + 2$, replacing them by $a - 1$ and $b + 1$ changes their squared sum by $-2(a - b - 1) < 0$; since each exchange strictly decreases the sum and there are only finitely many partitions of d into ℓ positive parts, iteration must terminate at a partition with no two blocks differing by more than 1, i.e. the balanced partition. ☐

Lemma 5.13 records the minimum $Q_\ell(d)$ used in the pinching thresholds; balanced blocks minimise $\sum_j b_j^2$ at fixed ℓ.

Definition 5.14 (Certified instrument upper envelope). *For $0 \le r \le D^{(n)}_{A,B}$, let $U^{(n)}_r(A,B)$ be the minimum of*

$$2\Big(1 - \frac{1}{nd_B \min\{d_A, d_B\}}\Big), \qquad \frac{2(D^{(n)}_{A,B} - r)}{D^{(n)}_{A,B} - r + 1},$$

all output-pinching values $2(1-1/\ell_{\rm out})$ satisfying $r \ge d_A^2(nQ_{\ell_{\rm out}}(d_B)-1)$, and all input-pinching values $2(1-1/k_{\rm in})$ satisfying $r \ge Q_{k_{\rm in}}(d_A)(nd_B^2-1)$. Here $\ell_{\rm out}$ ranges over $1,\dots,d_B$ and $k_{\rm in}$ over $1,\dots,d_A$, these being exactly the block counts of admissible decompositions into positive-dimensional blocks; families outside these ranges are ignored, and an empty pinching family is ignored.

Proposition 5.15 (Error-one floor of the certified upper envelope). *For $nd_B > 1$ and $0 \le r < D^{(n)}_{A,B}$, every admissible upper-bound term is at least 1, hence $U^{(n)}_r(A,B) \ge 1$.*

Proof. The constant-code bound satisfies $R_n = 2(1 - 1/(nd_Bs)) \ge 1$ because $nd_Bs \ge 2$. The convex-projection bound satisfies $2(D-r)/(D-r+1) \ge 1$ because $D - r \ge 1$, with equality only at $r = D - 1$. Output pinching gives $2(1 - 1/\ell_{\rm out}) \ge 1$ whenever $\ell_{\rm out} \ge 2$; $\ell_{\rm out} = 1$ requires $r \ge d_A^2(nQ_1(d_B) - 1) = d_A^2(nd_B^2 - 1) = D$. Similarly $k_{\rm in} \ge 2$ below D, because $Q_1(d_A)(nd_B^2 - 1) = D$. ☐

Corollary 5.16 (Exact constant-code instrument width). *Assume $nd_B > 1$. Then*

$$\delta^{\diamond}_{0,\rm inst}(A,B) = R_{\rm inst}(\mathcal{I}_*) = 2\Big(1 - \frac{1}{nd_Bs}\Big).$$

Proof. The lower bound in Theorem 3.8 holds even for centres Ψ in the full affine space $\mathfrak{A}^{(n)}_{A,B}$, and the optimal centre $\mathcal{I}_*$ is itself an instrument, so the infimum over instruments equals the covering radius. ☐

Corollary 5.17 (Exact top-subcritical replacement-instrument width). *Assume $nd_B > 1$. Then*

$$\delta^{\rm c}_{nd_B^2-2}(\mathrm{RepInst}_n(A,B); \mathrm{RepInst}_n(A,B)) = 1.$$

When $d_A = 1$ the same equality holds for the full instrument body.

Proof. Compose any continuous encoder into $\mathbb{R}^{nd_B^2-2}$ with the perfectly separated sphere of Theorem 4.5; Borsuk–Ulam and the triangle inequality give the lower bound 1. The replacement body is compact, convex, of affine dimension $m = nd_B^2 - 1$ and diameter two; at the top subcritical index $r = m - 1$ the convex-projection bound of Proposition 5.8 — whose proof uses only compactness, convexity, affine dimension, and the diameter, hence applies verbatim to the replacement body in place of the full instrument body — gives

$$\frac{2(m-r)}{m-r+1} = \frac{2}{2} = 1,$$

so the upper bound matches. For $d_A = 1$ every instrument is a replacement instrument. ☐

Theorem 5.18 (Exact collapse when $nd_Bs = 2$). *Assume $nd_B > 1$ and $nd_B \min\{d_A, d_B\} = 2$. Then*

$$\delta^{\diamond}_{r,\rm inst}(A,B) = \begin{cases} 1, & 0 \le r < D^{(n)}_{A,B}, \\ 0, & r \ge D^{(n)}_{A,B}. \end{cases}$$

This occurs for (i) arbitrary-input two-outcome POVMs ($d_B = 1$, $n = 2$, any d_A), where $D^{(2)}_{A,1} = d_A^2$ (see Figure 6(c), which plots case (i) with $d_A = 2$), and (ii) trivial-input qubit channels ($d_A = 1$, $n = 1$, $d_B = 2$), where $D^{(1)}_{1,2} = 3$. Thus continuous compression of all two-outcome POVMs is solved exactly: error 1 until full affine dimension, then 0.

Proof. If $nd_B s = 2$, then $\rho_n = 1$ and $R_n = 1$. For every $r < D^{(n)}_{A,B}$, choose a subspace $W \subseteq \mathfrak{V}^{(n)}_{A,B}$ of dimension $r+1$ and define on the auxiliary Euclidean sphere $S^r \subset W$ the radial map

$$h(u) = \mathcal{I}_* + \rho_n \frac{u}{\|u\|_{\diamond,\mathrm{inst}}}.$$

Because $\rho_n = 1$ and Theorem 3.3 applies, $h(u)$ is an instrument for every u, and $\|h(u) - h(-u)\|_{\diamond,\mathrm{inst}} = 2\rho_n = 2$. By Borsuk–Ulam every continuous encoder to $\mathbb{R}^r$ and every decoder must produce error at least 1. Conversely, the constant-code bound gives $\delta^{\diamond}_{0,\mathrm{inst}} \le R_n = 1$, and by monotonicity of the width the value is at most 1 for all r. Hence the width is exactly 1 for $0 \le r < D^{(n)}_{A,B}$, and 0 at and above $D^{(n)}_{A,B}$ by Theorem 5.7. $\square$

Proposition 5.19 (Gap structure and endpoint exactness)**.** *Let $m_n = nd_B^2 - 1$, $D_n = D^{(n)}_{A,B}$, $\rho_n = 2/(nd_B s)$, $R_n = 2 - \rho_n$. Then:*

1. *At $r = 0$ the width is exactly R_n (Corollary 5.16), and Definition 4.17 sets $\beta_0^{\mathrm{best},(n)} = R_n$; the envelopes therefore coincide at $r = 0$, the antipodal one-sphere certificate alone leaving the residual $R_n - 1 = 1 - \rho_n$.*
2. *For $0 < r \le L^{(n)}_{A,B}$, $\beta_r^{\mathrm{best},(n)} = 1$ and $U_r^{(n)} \ge 1$; the true width satisfies $1 \le \delta^{\diamond}_{r,\mathrm{inst}} \le U_r^{(n)}$ and is not determined except in the cases below.*
3. *For $L^{(n)}_{A,B} < r < D_n$, $\beta_r^{\mathrm{best},(n)} = \gamma^{(n)}_{A,B}$ and $U_r^{(n)} \ge 1$, so the* certified *bracket has length at least $1 - \gamma^{(n)}_{A,B}$; this bounds the certificate, not the unknown true width.*
4. *For $r \ge D_n$ the width is exactly 0.*
5. *For the replacement subfamily, $\delta^{\mathrm{c}}_{m_n-1}(\mathrm{RepInst}_n; \mathrm{RepInst}_n) = 1$ exactly, and this equals the full body when $d_A = 1$.*
6. *When $nd_B s = 2$ the full width collapses to the step function of Theorem 5.18.*

Proof. Item 1: at $r = 0$ no pinching term is admissible — the one-block thresholds equal $D_n > 0$, the output thresholds for $\ell_{\mathrm{out}} \ge 2$ are at least $d_A^2(2n-1) \ge 1$, and the input thresholds for $k_{\mathrm{in}} \ge 2$ are at least $2(nd_B^2 - 1) \ge 2$ — so U_0 is the minimum of R_n and $2D_n/(D_n+1)$, and

$$2 - \frac{2}{D_n + 1} \ge 2 - \frac{2}{nd_B s} = R_n$$

because $D_n + 1 = d_A^2(nd_B^2 - 1) + 1 \ge nd_B^2 \ge nd_B s$, using $d_A \ge 1$ and $d_B \ge s$; hence $U_0 = R_n$; with the upgraded value $\beta_0^{\mathrm{best},(n)} = R_n$ (Definition 4.17) the two envelopes coincide at $r = 0$, the one-sphere certificate alone leaving the residual $R_n - 1 = 1 - \rho_n$. Items 2–6 follow from the bracket (Corollary 5.4), the floor (Proposition 5.15), Corollaries 5.16 and 5.17, and Theorem 5.18. $\square$

Corollary 5.20 (Asymptotic positivity–dimension product)**.**

$$\Pi_n := D^{(n)}_{A,B}\, \rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) = d_A^2(nd_B^2 - 1) \frac{2}{nd_B \min\{d_A, d_B\}}$$
$$\longrightarrow\ \Pi_\infty := \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} = 2d_A \max\{d_A, d_B\}$$

as $n \to \infty$. Thus the product has a finite large-n limit despite the linear growth of the affine dimension and the $1/n$ decay of the centred radius; the convergence is from below, with the exact finite-n formula displayed in the proof below.

Proof. The limit follows from the displayed explicit formula:

$$\Pi_n = \frac{2d_A^2(nd_B^2 - 1)}{nd_B \min\{d_A, d_B\}} = \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} - \frac{2d_A^2}{nd_B \min\{d_A, d_B\}}$$
$$\xrightarrow[n\to\infty]{} \frac{2d_A^2 d_B}{\min\{d_A, d_B\}} = 2d_A \max\{d_A, d_B\}.$$

□

Corollary 5.21 (Scaling with the number of outcomes)**.** *Fix d_A, d_B and consider latent dimensions r_n with $r_n/n \to c$.*

1. *If $c < d_B^2$, then $r_n \le nd_B^2 - 2 \le L_{A,B}^{(n)}$ eventually, so $\beta_{r_n}^{\mathrm{best},(n)} = 1$ for all large n with $r_n \ge 1$, while $r_n = 0$ carries the exact value $R_n \to 2$: the certified lower bound saturates at its maximal value 1 already through flagged output preparation (in particular, for every fixed $r \ge 1$ one has $\delta_{r,\mathrm{inst}}^{\diamond} \ge 1$ for all large n, and $\delta_{0,\mathrm{inst}}^{\diamond} = R_n \ge 1$ exactly).*
2. *If $d_B^2 < c < d_A^2 d_B^2$, then eventually $L_{A,B}^{(n)} < r_n < D_{A,B}^{(n)}$, since $L_{A,B}^{(n)} = nd_B^2 + O(1)$ and $D_{A,B}^{(n)} = nd_A^2 d_B^2 + O(1)$. The certified lower bound is then $\beta_{r_n}^{\mathrm{best},(n)} = \gamma_{A,B}^{(n)} \to 1/d_A$, an n-independent constant strictly stronger than the $1/n$-decaying ball bound, while every term of the certified upper envelope (Definition 5.14) remains at least 1; the certified bracket therefore has asymptotic length at least $1 - 1/d_A$ in this regime.*
3. *If $c > d_A^2 d_B^2$, then $r_n \ge D_{A,B}^{(n)}$ eventually and $\delta_{r_n,\mathrm{inst}}^{\diamond} = 0$ exactly.*

The boundary cases $c = d_B^2$ and $c = d_A^2 d_B^2$ are left to a more refined analysis. Thus the number of outcomes linearly enlarges the parameter count needed for exact reconstruction while dividing the centred positivity margin by n, and the intermediate obstruction carries a constant, n-independent component $1/d_A$.

Proof. The asymptotic comparisons $L_{A,B}^{(n)} = nd_B^2 + O(1)$ and $D_{A,B}^{(n)} = nd_A^2 d_B^2 + O(1)$ follow from Definitions 4.17 and 5.14 together with $L_{\mathrm{rep}} = nd_B^2 - 2$, $L_{\mathrm{obs}} = d_A^2 - 2$, and $L_{\mathrm{join}} = nd_B^2 + O(1)$ when valid; they determine, by comparison with $r_n/n \to c$, which of the three ranges of the envelope contains r_n for all large n. The displayed values are then the piecewise definition of $\beta_r^{\mathrm{best},(n)}$ (Definition 4.17) and the exact threshold of Theorem 5.7. □

Remark 5.22 (Error-one barrier, scoped)**.** *For $r < D_{A,B}^{(n)}$ every admissible term of the certified upper envelope (Definition 5.14) is at least 1 (Proposition 5.15). Consequently the three constructions — constant codes, linear/convex projection with Helly selection, and balanced block pinching — cannot achieve error $\varepsilon < 1$ in the intermediate regime $L_{A,B}^{(n)} < r < D_{A,B}^{(n)}$. The bound therefore concerns these specific constructions rather than all possible codes; nonlinear manifold encodings or other decoder families are not excluded.*

It is natural to ask whether either of two further tools lowers the certified upper bounds below one in the intermediate regime; neither does. (i) Covering-number encodings. *If the body contains a diamond ball of radius r_0 (which exists by full affine dimension and norm equivalence), then an ε-net of the body in the instrument diamond norm has size at least $(r_0/\varepsilon)^D$ by the volume estimate; a net-based code thus needs at least $D \log_2(r_0/\varepsilon)$ bits, exceeding D bits for every $\varepsilon < r_0/2$. The affine encoding achieves error 0 with D Euclidean coordinates, and Theorem 5.7 shows that D coordinates are also necessary among continuous encoders; this comparison with discrete net-based codes is informal and does not by itself rule out other continuous low-dimensional encodings. (ii)* Orthogonal joins of full-body spheres. *These are precisely the constructions already absorbed into $\beta_r^{\mathrm{best},(n)}$ in Section 4: within this family they certify the value 1 up to $L_{A,B}^{(n)}$, and no join of replacement-type spheres extends the regime of* *antipodal separation 2 past the join dimension of Corollary 4.14; whether constructions outside this family extend the regime is exactly open problem (ii) of Section 7. The complete picture is summarised in Table 1.*

6 Specialisations, examples, and explicit trade-off

The following explicit consequences summarise the main results at the boundaries of the parameter space — channels ($n = 1$), POVMs ($d_B = 1$), and input-independent instruments ($d_A = 1$) — together with the explicit trade-off between outcome number and local robustness and the worked example used for the figures and tables.

- **Channel case $n = 1$.** The instrument body reduces to the channel body, and Theorem 3.4 gives, for $d_B > 1$,
$$\rho_\diamond(\Omega_{A,B}) = \frac{2}{d_B \min\{d_A, d_B\}},$$
with covering radius $2(1 - 1/(d_B \min\{d_A, d_B\}))$ and zero-error threshold $d_A^2(d_B^2 - 1)$. For $d_B = 1$ the channel body is the singleton trace functional of Remark 2.5. (All normalisations are those of Remark 2.1.)

- **POVM case $d_B = 1$, $n \geq 2$.** Every instrument has the form $\mathcal{I}_k(X) = \mathrm{Tr}(E_k X)$ with $E_k \geq 0$, $\sum_k E_k = I_A$ (Remark 2.4). The centred in-radius is $\rho_{\diamond,\mathrm{inst}}(\mathcal{I}_*) = 2/n$, the replacement sphere has dimension $n - 2$, and the affine dimension is $d_A^2(n - 1)$.

- **Input-independent case $d_A = 1$.** The map
$$\mathcal{I}_k(x) = x\,\tau_k, \qquad x \in \mathbb{C}, \qquad \tau \in \mathcal{S}_{\mathrm{flag}}(B \otimes C_n),$$
is an isometry $\mathrm{Inst}_n(\mathbb{C}, B) \cong \mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ with
$$\|\vec{\mathcal{R}}_\tau - \vec{\mathcal{R}}_\sigma\|_{\diamond,\mathrm{inst}} = \sum_k \|\tau_k - \sigma_k\|_1,$$
because the input is one-dimensional. Hence the state-space results for $\mathcal{S}_{\mathrm{flag}}(B \otimes C_n)$ transfer verbatim — the affine dimension, the replacement profile, the top-subcritical width (Corollary 5.17), the zero-error threshold, and the collapse case when $nd_B = 2$ — and the full-body antipodal profile is exactly
$$\alpha_r(\mathrm{Inst}_n(\mathbb{C}, B)) = \begin{cases} 1, & 0 \leq r \leq nd_B^2 - 2, \\ 0, & r \geq nd_B^2 - 1. \end{cases}$$
The centred in-radius is still $2/(nd_B)$ (since $s = \min\{1, d_B\} = 1$) as a radius of the body inside its affine hull, but it plays no role in the antipodal profile: for $d_A = 1$ the replacement body is the whole body (the isometry above identifies them), so the profile is governed by the full body's own Jordan construction.

- **$1/n$ suppression versus n-fold growth.** The centred in-radius at the uniform centre scales as $1/n$ for fixed d_B, while the affine dimension $D_{A,B}^{(n)} = d_A^2(nd_B^2 - 1)$ grows linearly in n: more outcomes increase the parameter count for exact reconstruction but reduce local robustness, with the precise trichotomy of Corollary 5.21.

The subcritical widths of the input-independent body are treated in the results below. The obstructions are Theorem 6.1 (an elementary median matching bound at $r = 1$, upgraded from the earlier five-outcome counterexample to the coordinate flags (j, k) and the constant

$2(q-1)/q$), Theorem 6.2 (a topological Tverberg bound at every $r \geq 2$), and Theorem 6.13, an equal-value-basis theorem for the pure states of a three-dimensional system, proved by a Chern-class obstruction on the quotient of the flag manifold under the cyclic permutation of its lines; the first two bound every subcritical width away from 1 as soon as $nd_B \geq 2r+3$ (Corollary 6.4), while the third closes the bottom index completely through the state-space bound of Theorem 6.14 and the classification of Corollary 6.16. The constructive side consists of the pair-mass and mass-split encoders (Propositions 6.6 and 6.5) and the flatness Theorems 6.8 and 6.10, which combine with the obstructions into the complete dichotomies of Corollaries 6.9 and 6.11; Theorem 6.12 determines the width at $r = n-1$ exactly for prime-power output dimensions, and Conjecture 6.19 states the remaining open direction.

Theorem 6.1 (Non-flat bottom width for $d_A = 1$: the median matching bound)**.** *Assume $d_A = 1$ and let $q \geq 2$ be an integer with $2q-1 \leq nd_B$. Then*

$$\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) \;\geq\; \frac{2(q-1)}{q}.$$

In particular $\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) \geq 4/3 > 1$ whenever $nd_B \geq 5$ — for every $n \geq 1$ and every $d_B \geq 1$, including the one-outcome state space when $d_B \geq 5$ — and, with q maximal, $q = \lfloor (nd_B+1)/2 \rfloor$,

$$\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) \;\geq\; 2\,\frac{\lfloor (nd_B+1)/2 \rfloor - 1}{\lfloor (nd_B+1)/2 \rfloor}.$$

For $q = 2$ the bound is the antipodal floor $\delta_1 \geq 1$.

Proof. Under the identification of the input-independent case, write instruments as n-tuples $\tau = (\tau_1, \ldots, \tau_n)$ of positive blocks with $\sum_k \operatorname{Tr} \tau_k = 1$, normed by $\|\tau - \sigma\|_{\diamond,\mathrm{inst}} = \sum_k \|\tau_k - \sigma_k\|_1$. Fix an orthonormal basis $(\psi_1, \ldots, \psi_{d_B})$ of B (for $d_B = 1$, $\psi_1 = 1$). For a coordinate $u = (j,k) \in [n] \times [d_B]$ let w_u be the instrument whose j-th block is $\psi_k \psi_k^*$ and whose other blocks vanish; these nd_B *coordinate flags* satisfy $\|w_u - w_{u'}\| = 2$ for $u \neq u'$. For an instrument $c = (c_1, \ldots, c_n)$ define the *coordinate weights* $c_u := \langle \psi_k | c_j | \psi_k \rangle \geq 0$ for $u = (j,k)$; then $\sum_u c_u = \sum_j \operatorname{Tr} c_j = 1$. Two bounds hold for every instrument c.

1. *Vertex bound:* $\|w_u - c\| \geq 2(1 - c_u)$. Indeed $X := \psi_k \psi_k^* - c_j$ is Hermitian and traceless, so $\|X\|_1 = 2 \operatorname{Tr} X_+ \geq 2\langle \psi_k | X | \psi_k \rangle = 2(1 - c_u)$, where X_+ is the positive part; the remaining blocks contribute $\sum_{j' \neq j} \|c_{j'}\|_1 = 1 - \operatorname{Tr} c_j \geq 0$.
2. *Edge bound:* for $z = (1-\lambda) w_u + \lambda w_v$ with $u \neq v$ and $\lambda \in [0,1]$, $\|z - c\| \geq 2(1 - c_u - c_v)$. For Hermitian X and Hermitian Y with $\|Y\|_\infty \leq 1$ one has $\|X\|_1 \geq |\operatorname{Tr}(XY)|$ (trace-norm duality); taking $Y_j = \sum_k \varepsilon_{jk} |\psi_k\rangle\langle\psi_k|$ with signs ε aligned with the entries gives

$$\|z - c\| = \sum_j \|z_j - c_j\|_1 \;\geq\; \sum_{u'} \big|\langle \psi_{k'} | z_{j'} - c_{j'} | \psi_{k'} \rangle\big| \;\geq\; \big|1 - c_u - c_v\big| + \sum_{u' \notin \{u,v\}} c_{u'} \;\geq\; 2(1 - c_u - c_v),$$

because the coordinate entries of z are $1-\lambda$ at u, λ at v and 0 elsewhere, and $c_u + c_v \leq 1$. The same computation shows that *any* instrument x whose coordinate entries vanish off a coordinate set V — in particular any convex combination of flags with coordinates in V — satisfies $\|x - c\| \geq 2(1 - \sum_{u \in V} c_u)$, since $\sum_{u \in V} x_u = 1$.

Let $f : \mathrm{Inst}_n(\mathbb{C}, B) \to \mathbb{R}$ be a continuous encoder and $g : \mathbb{R} \to \mathrm{Inst}_n(\mathbb{C}, B)$ a decoder, and let t^* be the q-th smallest of the nd_B values $f(w_u)$. Write $G = \{u : f(w_u) = t^*\}$, $P = \#\{u : f(w_u) < t^*\}$ and $Q = \#\{u : f(w_u) > t^*\}$. By the choice of t^*, $P \leq q-1$ and $P + |G| \geq q$; since $nd_B \geq 2q-1$,

$$Q \;\geq\; nd_B - P - |G| \;\geq\; (2q-1) - (q-1) - |G| \;=\; q - |G|.$$

Choose a bipartite matching of size $m := \min\{P, Q\} \geq q - |G|$ between the flags below and the flags above t^* (a matching saturating the smaller side). For every matched pair (b, a) one has $f(w_b) < t^* < f(w_a)$, so by the intermediate value theorem the segment $[w_b, w_a]$ contains a point z_{ba} with $f(z_{ba}) = t^*$. The leaf $f^{-1}(t^*)$ therefore contains the $|G|$ flags $\{w_u\}_{u\in G}$ and the m crossing points: $|G|+m \geq q$ points whose coordinate supports — singletons $\{u\}$ and pairs $\{b, a\}$ — are pairwise disjoint. Let $V_1, \dots, V_{|G|+m}$ be those supports and x_i the corresponding points. The decoder value $c = g(t^*)$ satisfies $\sum_i \sum_{u\in V_i} c_u \leq \sum_u c_u = 1$, so some i has $\sum_{u\in V_i} c_u \leq 1/q$, and the bounds above give $\|x_i - c\| \geq 2(1 - 1/q)$: every encoder/decoder pair suffers worst-case error at least $2(q-1)/q$, so the infimum over such pairs is at least $2(q-1)/q$. □

Theorem 6.2 (Topological Tverberg bound, every latent dimension)**.** *Assume $d_A = 1$ and let $q \geq 2$ be a prime power with $(q-1)(r+1) \leq nd_B - 1$. Then*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) \;\geq\; \frac{2(q-1)}{q}.$$

Proof. Retain the coordinate flags w_u, the coordinate weights c_u, and the supporting bound for instruments supported on a coordinate set from the proof of Theorem 6.1. The affine map $\Phi : x \mapsto \sum_u x_u w_u$ embeds the simplex Σ_{nd_B-1} into $\mathrm{Inst}_n(\mathbb{C}, B)$: the j-th block of $\Phi(x)$ is $\sum_k x_{(j,k)}\psi_k\psi_k^*$, which determines x, and the face of Σ_{nd_B-1} on a vertex set V is mapped into the instruments supported on V. Let $f : \mathrm{Inst}_n(\mathbb{C}, B) \to \mathbb{R}^r$ be a continuous encoder and $g : \mathbb{R}^r \to \mathrm{Inst}_n(\mathbb{C}, B)$ a decoder, and set $F := f \circ \Phi$, a continuous map $\Sigma_{nd_B-1} \to \mathbb{R}^r$. Since $(q-1)(r+1) \leq nd_B - 1$, the simplex has a face Σ' on $(q-1)(r+1)+1$ vertices; restrict F to Σ'. By the topological Tverberg theorem in the prime-power case [17] (for $q = 2$ it is the topological Radon theorem of Borsuk–Ulam type [15], and for affine maps on vertex images it reduces to Tverberg's partition theorem [18]), there are q pairwise vertex-disjoint faces $F_1, \dots, F_q$ of Σ' whose images under F share a point y. Pick $x_i \in F_i$ with $F(x_i) = y$. The instruments $\Phi(x_i)$ lie in the leaf $f^{-1}(y)$ and are supported on the pairwise disjoint vertex sets V_i of F_i. For $c = g(y)$, $\sum_i \sum_{u\in V_i} c_u \leq 1$, so some i has $\sum_{u\in V_i} c_u \leq 1/q$, whence

$$\|g(f(\Phi(x_i))) - \Phi(x_i)\| \;\geq\; 2\Big(1 - \frac{1}{q}\Big),$$

and every encoder/decoder pair suffers worst-case error at least $2(q-1)/q$. □

Remark 6.3 (Prime powers)**.** *The prime-power hypothesis is what the topological Tverberg theorem provides. At $r = 1$ Theorem 6.1 is purely combinatorial and carries the same constant $2(q-1)/q$ for* every *$q \leq (nd_B+1)/2$, prime power or not. The applications below use $q \in \{2, 3\}$ and, in Theorem 6.12, $q = d_B$ itself when that is a prime power.*

Corollary 6.4 (The flat value fails below the diagonal $nd_B = 2r + 2$)**.** *Assume $d_A = 1$, $r \geq 1$, and $nd_B \geq 2r + 3$. Then*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) \;\geq\; \frac{4}{3}\ (> 1).$$

In particular the flat value 1 fails at $r = 2$ already for $(n, d_B) \in \{(7, 1), (2, 4)\}$: flatness throughout $2 \leq r \leq nd_B^2 - 2$ is refuted at every output dimension, and the corrected statement is Conjecture 6.19.

Proof. For $r = 1$ apply Theorem 6.1 with $q = 3$, which requires $2q - 1 = 5 \leq nd_B$. For $r \geq 2$ apply Theorem 6.2 with the prime $q = 3$, which requires $(3-1)(r+1) = 2r + 2 \leq nd_B - 1$, i.e. $nd_B \geq 2r + 3$. □

Proposition 6.5 (Pair-mass and mass-split encoders)**.** *Assume $d_A = 1$.*

1. Pair-mass reduction (classical outputs). *Let $d_B = 1$ and $n \geq 3$. Under the identification of the input-independent case with the simplex $\Sigma_{n-1} = \{x \in \mathbb{R}^n_{\geq 0} : \sum_k x_k = 1\}$ in the ℓ_1 norm,*

suppose the $(n-2)$-outcome simplex Σ_{n-3} carries a continuous encoder $\varphi : \Sigma_{n-3} \to \mathbb{R}^{r-1}$ with decoder ψ and worst-case error E. Then Σ_{n-1} carries the r-dimensional pair

$$f(x) = \big(x_1 + x_2,\ m\,\varphi(\tilde{x})\big), \qquad g(t,w) = \Big(\frac{t}{2}, \frac{t}{2},\ m\,\psi\big(\tfrac{w}{m}\big)\Big),$$

where $m = 1 - x_1 - x_2$ and $\tilde{x} = (x_3, \dots, x_n)/m$ (at $m = 0$ set the second coordinate of f and the tail of g to 0; f is continuous because φ is bounded), with worst-case error at most $\max\{1, E\}$.

2. Mass-split encoder (every output dimension). *Let $d_B \geq 2$ and $n \geq 2$. Then* $\mathrm{Inst}_n(\mathbb{C}, B)$ *carries an $(n-1)$-dimensional encoder/decoder pair with worst-case error at most $2(1 - \frac{1}{d_B})$: encode the $n-1$ outcome traces,*

$$f(\tau) = \big(\operatorname{Tr}\tau_1, \dots, \operatorname{Tr}\tau_{n-1}\big),$$

and decode block $k < n$ as $t_k I_B/d_B$ and the last block as $\big(1 - \sum_{k<n} t_k\big)\, I_B/d_B$, where t_k is the k-th latent coordinate. Appending zero coordinates extends the pair to every $r \geq n-1$.

Proof. (i) Fix $x \in \Sigma_{n-1}$ and write $t = x_1 + x_2$. The reconstruction error splits as

$$\|x - g(f(x))\|_1 = \Big|x_1 - \frac{t}{2}\Big| + \Big|x_2 - \frac{t}{2}\Big| + \big\|x_{\text{tail}} - m\,\psi(\varphi(\tilde{x}))\big\|_1 \leq t + mE \leq \max\{1, E\},$$

using $2|x_1 - t/2| \leq t$ (as $0 \leq x_1 \leq t$), the homogeneity $\|m\tilde{x} - m\psi\|_1 = m\|\tilde{x} - \psi\|_1$, and $t + m = 1$; at $m = 0$ the tail error vanishes and $t = 1$. (ii) A block τ_k of trace $t_k > 0$ normalises to the state $\rho = \tau_k/t_k$, and $\|\rho - I_B/d_B\|_1 = 2\big(1 - \sum_i \min\{\lambda_i, 1/d_B\}\big) \leq 2(1 - 1/d_B)$ because the largest eigenvalue of ρ is at least $1/d_B$; a block of trace zero vanishes. The decoder's last-block trace $1 - \sum_{k<n} t_k$ equals $\operatorname{Tr}\tau_n$ exactly, so every block is decoded within $2(1 - 1/d_B)$ times its own trace, and the traces sum to one: the total error is at most $2(1 - 1/d_B)$. The decoder is affine on the image of f, the compact simplex $\{t \in \mathbb{R}^{n-1}_{\geq 0} : \sum_{k<n} t_k \leq 1\}$, and is extended arbitrarily off the image, which Definition 5.2 permits. □

Proposition 6.6 (Pair-mass encoder at the bottom index)**.** *Assume $d_A = d_B = 1$ and $n \geq 3$. Then*

$$\delta^{\diamond}_{1,\text{inst}}(\mathbb{C}, \mathbb{C}) \ \leq\ \max\Big\{1,\ \frac{2(n-3)}{n-2}\Big\},$$

and the right-hand side equals $2(n-3)/(n-2)$ for $n \geq 5$ (and 1 for $n \in \{3, 4\}$).

Proof. Run the base construction of Theorem 6.8 at general n: the encoder $f(x) = x_1 + x_2$ with decoder

$$g(t) = \Big(\frac{t}{2}, \frac{t}{2}, \frac{1-t}{n-2}, \dots, \frac{1-t}{n-2}\Big) \in \Sigma_{n-1}.$$

For x with $x_1 + x_2 = t$ the error splits as

$$\|x - g(t)\|_1 = \Big|x_1 - \frac{t}{2}\Big| + \Big|x_2 - \frac{t}{2}\Big| + \Big\|x_{\text{tail}} - \frac{1-t}{n-2}\mathbf{1}\Big\|_1 \ \leq\ t + \frac{2(n-3)}{n-2}\,(1-t),$$

using $2|x_1 - t/2| \leq t$ and the worst deviation $2(n-3)/(n-2)$ of a distribution of mass $1-t$ on $n-2$ coordinates from the uniform one (attained by concentration). The right-hand side is affine in t with slope $(4-n)/(n-2)$, so its maximum over $t \in [0,1]$ is $\max\{1, 2(n-3)/(n-2)\}$, attained at $t = 1$ for $n = 3$ and at $t = 0$ for $n \geq 4$. □

Corollary 6.7 (Exact bottom and boundary widths)**.** *Assume $d_A = d_B = 1$.*

1. *At $n = 5$: $\delta^{\diamond}_{1,\text{inst}}(\mathbb{C}, \mathbb{C}) = 4/3$ exactly (the four-dimensional simplex).*
2. *More generally, $\delta^{\diamond}_{r,\text{inst}}(\mathbb{C}, \mathbb{C}) = 4/3$ exactly at $n = 2r + 3$, for every $r \geq 1$.*

Proof. (1) Theorem 6.1 with $q = 3$ gives $\delta_1 \geq 4/3$ at $n = 5$, and Proposition 6.6 gives $\delta_1 \leq 2(5-3)/(5-2) = 4/3$. (2) Corollary 6.4 gives $\delta_r \geq 4/3$ at $n = 2r+3$; for the upper bound, iterate the pair-mass reduction (Proposition 6.5(i)) from $(n, r) = (2r+3, r)$ down to $(5, 1)$, each step preserving error at most $\max\{1, E\}$, so $\delta_r \leq \max\{1, \delta_1(\Sigma_4)\} = 4/3$. □

Theorem 6.8 (Flat classical widths on the upper half of the subcritical range)**.** *Assume $d_A = d_B = 1$ and $n \geq 3$. Then*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, \mathbb{C}) = 1 \qquad \textit{for} \qquad \left\lceil \frac{n-2}{2} \right\rceil \leq r \leq n-2.$$

Proof. The lower bound $\delta_r \geq 1$ holds on the whole range $1 \leq r \leq n-2$ by the exact antipodal profile of the input-independent case displayed above. For the upper bound, Proposition 6.5(i) reduces the pair (n, r) to $(n-2, r-1)$ at error at most 1 whenever the smaller problem has error at most 1; iterating reaches the base $r = 1$ exactly when $n \leq 2r+2$, that is, $r \geq \lceil (n-2)/2 \rceil$. It remains to verify the bases $n \in \{3, 4\}$ at $r = 1$: the one-dimensional encoder $f(x) = x_1 + x_2$ with decoder

$$g(t) = \left(\frac{t}{2}, \frac{t}{2}, \frac{1-t}{n-2}, \ldots, \frac{1-t}{n-2}\right) \in \Sigma_{n-1}$$

satisfies, for every x with $x_1 + x_2 = t$,

$$\sum_{k=1}^{n} \min\{x_k, g(t)_k\} \geq \frac{1}{2}, \qquad \text{hence} \qquad \|x - g(t)\|_1 = 2\Big(1 - \sum_k \min\{x_k, g(t)_k\}\Big) \leq 1:$$

indeed $\min\{x_1, t/2\} + \min\{x_2, t/2\} \geq t/2$, because the smaller of x_1, x_2 contributes itself and the larger at least $t/2$; for $n = 3$ the single tail coordinate satisfies $\min\{x_3, 1-t\} = 1-t$ exactly; and for $n = 4$ the same two-coordinate argument applied to $x_3 + x_4 = 1-t$ gives $\min\{x_3, (1-t)/2\} + \min\{x_4, (1-t)/2\} \geq (1-t)/2$. □

Corollary 6.9 (Classical dichotomy)**.** *Assume $d_A = d_B = 1$ and $n \geq 3$. On the subcritical range $1 \leq r \leq n-2$,*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, \mathbb{C}) = 1 \qquad \Longleftrightarrow \qquad n \leq 2r+2,$$

and $\delta_r \geq 4/3$ for $n \geq 2r+3$.

Proof. Combine Theorem 6.8 (flatness for $r \geq \lceil (n-2)/2 \rceil$, i.e. $n \leq 2r+2$) with Corollary 6.4. □

Theorem 6.10 (Flat qubit-output widths from latent dimension $n-1$)**.** *Assume $d_A = 1$, $d_B = 2$, and $n \geq 2$. Then*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) = 1 \qquad \textit{for} \qquad n-1 \leq r \leq 4n-2.$$

Proof. The lower bound is again the exact antipodal profile, valid through $r \leq n d_B^2 - 2 = 4n-2$. For the upper bound, the mass-split encoder of Proposition 6.5(ii) at $d_B = 2$ has worst-case error at most $2(1 - 1/2) = 1$ at latent dimension $r = n-1$; appending zero coordinates extends it to every $r \geq n-1$. □

Corollary 6.11 (Qubit dichotomy)**.** *Assume $d_A = 1$, $d_B = 2$, and $n \geq 2$. On the subcritical range $1 \leq r \leq 4n-2$,*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) = 1 \quad \Longleftrightarrow \quad r \geq n-1, \qquad \textit{and} \qquad \delta_r \geq 4/3 \textit{ for } r \leq n-2.$$

Proof. Theorem 6.10 gives the flat direction; Corollary 6.4 with $n d_B = 2n \geq 2r+3 \iff r \leq n-2$ gives the obstruction. □

Theorem 6.12 (Exact width at $r = n-1$ for prime-power output dimension)**.** *Assume $d_A = 1$, $n \geq 2$, and let $d_B \geq 2$ be a prime power. Then*

$$\delta^{\diamond}_{n-1,\mathrm{inst}}(\mathbb{C}, B) \;=\; 2\Big(1 - \frac{1}{d_B}\Big) \qquad \textit{exactly}$$

(for $d_B = 2$ this is the flat value of Theorem 6.10; for $d_B \geq 3$ it exceeds 1, so the subcritical range is non-flat at $r = n-1$).

Proof. The upper bound is the mass-split encoder of Proposition 6.5(ii). For the lower bound apply Theorem 6.2 with $q = d_B$ (a prime power by assumption): the hypothesis $(q-1)(r+1) = (d_B - 1)n \leq nd_B - 1$ holds for every $n \geq 1$, and $2(q-1)/q = 2(1 - 1/d_B)$. ☐

Theorem 6.13 (Equal-value orthonormal bases)**.** *Let V be a three-dimensional complex Hilbert space and let $f\colon \mathbb{P}(V) \to \mathbb{R}$ be continuous on the pure states. Then V admits an orthonormal basis (ψ_1, ψ_2, ψ_3) with*

$$f([\psi_1]) \;=\; f([\psi_2]) \;=\; f([\psi_3]).$$

Proof. Realise the manifold of complete flags of V as the closed set of ordered pairs of orthogonal lines,

$$\mathrm{Fl} = \big\{(\ell_1, \ell_2) \in \mathbb{P}(V) \times \mathbb{P}(V) : \ell_1 \perp \ell_2\big\}, \qquad \ell_3(\ell_1, \ell_2) := (\ell_1 \oplus \ell_2)^{\perp},$$

and let $c(\ell_1, \ell_2) = (\ell_2, \ell_3)$; then $c^3 = \mathrm{id}$, and c has no fixed point (a fixed point would force $\ell_2 = \ell_1$, impossible for orthogonal lines). With $p_1(\ell_1, \ell_2) = \ell_1$ put $g = f \circ p_1$. A point $x = (\ell_1, \ell_2)$ satisfies $g(x) = g(cx) = g(c^2x)$ exactly when f takes one common value on the three lines $\ell_1, \ell_2, \ell_3(x)$, which form an orthonormal basis of V; the theorem asserts that such an x exists.

Define $h = g - g \circ c$ and $H = (h,\ h \circ c)\colon \mathrm{Fl} \to \mathbb{R}^2$. The telescoping identity $h + h\circ c + h \circ c^2 \equiv 0$ (the sum collapses to $g - g \circ c^3 = g - g$) yields

$$H \circ c = T \circ H, \qquad T = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix},$$

where T has order three, determinant one, and $T \neq I$. The eigenvalues of T are the primitive cube roots of unity, so $T = SRS^{-1}$ with S real invertible and R the plane rotation by $2\pi/3$; replacing H by $S^{-1}H$ (which does not change where H vanishes) we may take $T = R$. Suppose, for contradiction, that H never vanishes. Then $\widehat{H} = H/\|H\|_2\colon \mathrm{Fl} \to S^1$ is well defined and satisfies

$$\widehat{H}(cx) = R\,\widehat{H}(x) = \omega\,\widehat{H}(x), \qquad \omega = e^{2\pi i/3},$$

viewing $S^1 \subset \mathbb{C}$ and R as multiplication by ω.

The cyclic group $\langle c\rangle \cong \mathbb{Z}/3$ acts freely on Fl, so $B := \mathrm{Fl}/\langle c\rangle$ carries the regular three-fold covering $\pi\colon \mathrm{Fl} \to B$ with deck group $\mathbb{Z}/3$. Let $L \to B$ be the complex line bundle obtained as the quotient of $\mathrm{Fl} \times \mathbb{C}$ by the equivalence $(x, z) \sim (cx, \omega z)$. Its sections are exactly the continuous $s\colon \mathrm{Fl} \to \mathbb{C}$ with $s \circ c = \omega s$, so $\widehat{H}$ is a nowhere-vanishing section of L; a complex line bundle with such a section is trivial, whence $c_1(L) = 0$ in $H^2(B; \mathbb{Z})$. We show $c_1(L) \neq 0$.

Let $\kappa\colon B \to B(\mathbb{Z}/3)$ classify the covering. The bundle L is the κ-pullback of the line bundle over $B(\mathbb{Z}/3)$ attached to the character $c \mapsto \omega$, whose first Chern class is the image of that character under the connecting homomorphism of $0 \to \mathbb{Z} \to \mathbb{R} \to U(1)$; the character is faithful, so this class generates $H^2(B(\mathbb{Z}/3); \mathbb{Z}) \cong \mathbb{Z}/3$ [19, 20]. Thus $c_1(L) = \kappa^*(\text{generator})$, and it suffices to prove that $\kappa^*\colon H^2(B(\mathbb{Z}/3); \mathbb{Z}) \to H^2(B; \mathbb{Z})$ is injective.

Consider the Cartan–Leray spectral sequence of the free cover π [19],

$$E_2^{p,q} = H^p\big(\mathbb{Z}/3;\ H^q(\mathrm{Fl}; \mathbb{Z})\big) \implies H^{p+q}(B; \mathbb{Z}).$$

Since $\mathrm{Fl} = U(3)/U(1)^3$ [21]: it is connected; it is simply connected, because in the homotopy exact sequence of $U(1)^3 \to U(3) \to \mathrm{Fl}$ the map $\pi_1(U(1)^3) = \mathbb{Z}^3 \to \pi_1(U(3)) = \mathbb{Z}$ is $(a, b, c) \mapsto a+b+c$ and is surjective (each phase loop winds the determinant once); its second cohomology is $H^2(\mathrm{Fl}; \mathbb{Z}) \cong \mathbb{Z}^3/\mathbb{Z}(1, 1, 1)$; and the deck transformation c, which cyclically permutes the three lines of every flag, acts on $H^2(\mathrm{Fl}; \mathbb{Z})$ by the cyclic permutation P of the three coordinates. The P-invariants of $\mathbb{Z}^3/\mathbb{Z}(1, 1, 1)$ vanish: from $Pv - v \in \mathbb{Z}(1, 1, 1)$ the three coordinates of v differ by one common constant k with $3k = 0$, hence $k = 0$ and $v \in \mathbb{Z}(1, 1, 1)$. Consequently

$$E_2^{1,1} = H^1(\mathbb{Z}/3; H^1(\mathrm{Fl}; \mathbb{Z})) = 0 \quad (H^1(\mathrm{Fl}; \mathbb{Z}) = 0), \qquad E_2^{0,2} = H^2(\mathrm{Fl}; \mathbb{Z})^{\mathbb{Z}/3} = 0,$$

so the only term of total degree two is $E_2^{2,0} = H^2(\mathbb{Z}/3; \mathbb{Z}) \cong \mathbb{Z}/3$. No differential touches it (for $r \geq 2$ the differential from $(2, 0)$ lands in $E_r^{2+r,1-r} = 0$, and the only possible incoming term $E_2^{0,1}$ vanishes since $H^1(\mathrm{Fl}) = 0$), so $E_\infty^{2,0} = \mathbb{Z}/3$ and, the other two graded pieces of total degree two being zero,

$$H^2(B; \mathbb{Z}) \;\cong\; \mathbb{Z}/3,$$

with the edge homomorphism $H^2(\mathbb{Z}/3; \mathbb{Z}) \to H^2(B; \mathbb{Z})$, which is κ^* [19], an isomorphism. Hence $c_1(L)$ is the generator of $H^2(B; \mathbb{Z}) \cong \mathbb{Z}/3$; in particular $c_1(L) \neq 0$, contradicting the triviality forced by the nowhere-vanishing section $\widehat{H}$.

So H vanishes at some $x \in \mathrm{Fl}$, that is, $h(x) = h(cx) = 0$ and therefore $g(x) = g(cx) = g(c^2x)$: the three lines of the flag x form an orthonormal basis of V on which f takes one common value. □

Theorem 6.14 (State-space bottom widths for $d_B \geq 3$)**.** *Assume $d_A = 1$ and $d_B \geq 3$. Then the one-outcome body — the state space $\mathcal{D}(B)$ under the identification of the input-independent case — satisfies*

$$\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) \;\geq\; \frac{4}{3},$$

with equality at $d_B = 3$; at $d_B = 4$ the width lies in $[\,4/3,\; 3/2\,]$.

Proof. Let $f\colon \mathrm{Inst}_1(\mathbb{C}, B) \to \mathbb{R}$ be a continuous encoder and $g\colon \mathbb{R} \to \mathrm{Inst}_1(\mathbb{C}, B)$ a decoder, and let $V \subseteq B$ be a three-dimensional subspace. The restriction of f to the pure states of V is continuous, so Theorem 6.13 supplies an orthonormal basis (ψ_1, ψ_2, ψ_3) of V with $f(\psi_i\psi_i^*) = t$ for $i = 1, 2, 3$. Write $c = g(t)$, an instrument of the one-outcome body, i.e. a state: $c \succeq 0$ and $\mathrm{Tr}\, c = 1$. Then

$$\sum_{i=1}^{3} \langle \psi_i | c | \psi_i \rangle = \mathrm{Tr}(c\, P_V) \;\leq\; \mathrm{Tr}\, c = 1,$$

because $c \succeq 0$ and $P_V \preceq I$; hence some i has $\langle \psi_i | c | \psi_i \rangle \leq \frac{1}{3}$. For that i the matrix $X = \psi_i\psi_i^* - c$ is Hermitian and traceless, so

$$\|X\|_1 = 2\; \mathrm{Tr}\, X_+ \;\geq\; 2\langle \psi_i | X | \psi_i \rangle = 2\Big(1 - \langle \psi_i | c | \psi_i \rangle\Big) \;\geq\; \frac{4}{3},$$

using $\langle \psi | X | \psi \rangle \leq \lambda_{\max}(X) \leq \mathrm{Tr}\, X_+$. Every encoder/decoder pair therefore suffers worst-case error at least $4/3$, which proves the lower bound (it is the vertex bound of Theorem 6.1, item 1, read on three flags). The upper bounds are the constant-code values of Corollary 5.16 at $n = 1$: $\delta_1 \leq \delta_0 = R_1 = 2(1 - 1/d_B)$, which equals $4/3$ at $d_B = 3$ and $3/2$ at $d_B = 4$. □

Remark 6.15 (Decoder scope of the state-space bound)**.** *At $d_B = 3$ the projection P_V is the identity, so $\sum_i \langle \psi_i | c | \psi_i \rangle = \mathrm{Tr}\, c = 1$ holds for every trace-one Hermitian c and the bound of Theorem 6.14 survives even decoders valued in the ambient affine space (Remark 5.3). At $d_B \geq 4$ the inequality $\mathrm{Tr}(cP_V) \leq \mathrm{Tr}\, c$ uses $c \succeq 0$, so the bound is proved for the width as defined, with instrument-valued decoders; an affine decoder can concentrate negative weight outside V, and Remark 5.3 does not extend to this argument.*

Corollary 6.16 (Complete bottom-width classification for $d_A = 1$)**.** *Assume $d_A = 1$ and $nd_B > 2$. Then*

$$\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) = 1 \qquad \Longleftrightarrow \qquad nd_B \le 4 \ \ \text{and} \ \ d_B \le 2,$$

and $\delta^{\diamond}_{1,\mathrm{inst}}(\mathbb{C}, B) \ge 4/3$ in every other case. The value $4/3$ is exact at the three instances $(n, d_B) = (5, 1)$, $(1, 3)$, and $(2, 3)$; the one-outcome ququart lies in $[4/3, 3/2]$, and so does the classical six-outcome simplex.

Proof. The flat cases with $nd_B \le 4$ and $d_B \le 2$ are $\{(2, 1), (3, 1), (4, 1)\}$ (the bases of Theorem 6.8), $(1, 2)$ (the collapse, Theorem 5.18), and $(2, 2)$ (the mass-split encoder, Proposition 6.5(ii) at $d_B = 2$). In every other case: $d_B = 1$ with $n \ge 5$, and $d_B = 2$ with $n \ge 3$, are covered by Theorem 6.1 with $q = 3$ ($nd_B \ge 5$); $n = 1$ with $d_B \ge 3$ is Theorem 6.14; $n \ge 2$ with $d_B \ge 3$ has $nd_B \ge 6 \ge 5$ and is covered by Theorem 6.1 again. The exact values are Corollary 6.7 at $(5, 1)$, Theorem 6.14 at $(1, 3)$, and Theorem 6.12 at $(2, 3)$ ($d_B = 3$ is a prime power); the stated brackets are the constant-code value at $(1, 4)$ and Proposition 6.6 at $n = 6$. ☐

Lemma 6.17 (Concentration gate)**.** *Assume $d_A = 1$ and $d_B \ge 2$. If an n-outcome instance is flat at some $r \ge 1$, $\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) = 1$, then the one-outcome body at the same output dimension satisfies $\delta^{\diamond}_{r,\mathrm{inst}}\big(\mathrm{Inst}_1(\mathbb{C}, B)\big) \le 1$. In particular every flat instance at output dimension d_B requires the state-space width $\delta_r\big(\mathcal{D}(B)\big) \le 1$ at the same latent dimension r.*

Proof. Let (f, g) achieve worst-case error at most 1 on $\mathrm{Inst}_n(\mathbb{C}, B)$ and let $\jmath\colon \mathrm{Inst}_1(\mathbb{C}, B) \to \mathrm{Inst}_n(\mathbb{C}, B)$ place the whole mass in the first slot, $\jmath(\rho) = (\rho, 0, \dots, 0)$; then $f \circ \jmath$ is a continuous encoder into $\mathbb{R}^r$. Write $g(t) = \big(g_1(t), \dots, g_n(t)\big)$ and fix a state $\sigma_0 \in \mathcal{D}(B)$. The blocks of $g(t)$ are positive semidefinite with $\sum_k \operatorname{Tr} g_k(t) = 1$, so the slack $1 - \operatorname{Tr} g_1(t) = \sum_{k \ge 2} \operatorname{Tr} g_k(t)$ is nonnegative, and the padded first block

$$\bar{g}_1(t) \ := \ g_1(t) + \big(1 - \operatorname{Tr} g_1(t)\big)\, \sigma_0$$

is again a state, so $\bar{g}_1 \colon \mathbb{R}^r \to \mathrm{Inst}_1(\mathbb{C}, B)$ is a decoder for the one-outcome body. For every $\rho \in \mathrm{Inst}_1(\mathbb{C}, B)$, using $\|Y\|_1 = \operatorname{Tr} Y$ for positive semidefinite Y,

$$\big\|\rho - \bar{g}_1(f(\jmath\rho))\big\|_1 \ \le \ \big\|\rho - g_1(f(\jmath\rho))\big\|_1 + \big(1 - \operatorname{Tr} g_1(f(\jmath\rho))\big) \ = \ \big\|\jmath(\rho) - g(f(\jmath\rho))\big\|_{\diamond,\mathrm{inst}} \ \le \ 1,$$

where the middle identity reassembles the block sum: the slack $\sum_{k \ge 2} \operatorname{Tr} g_k$ equals the tail contribution $\sum_{k \ge 2} \|g_k\|_1$. The padded pair $\big(f \circ \jmath,\ \bar{g}_1\big)$ is thus an encoder/decoder pair for the one-outcome body at latent dimension r with worst-case error at most 1, which proves the first claim; the second claim is the identification of $\mathrm{Inst}_1(\mathbb{C}, B)$ with $\mathcal{D}(B)$ carrying the trace norm. ☐

Remark 6.18 (Status of the flat-width problem for $d_A = 1$)**.** *The problem is governed by the flag-coordinate count nd_B, with a complete answer at the bottom index. On the obstruction side, the median matching bound at $r = 1$ (Theorem 6.1), the topological Tverberg bound at every $r \ge 2$ (Theorem 6.2), and the equal-value-basis theorem (Theorem 6.13) combine into Corollary 6.4: $\delta_r \ge 4/3$ whenever $nd_B \ge 2r + 3$, with the sharpening $\delta_r \ge 2(q - 1)/q$ available whenever $2q - 1 \le nd_B$ at $r = 1$ (all q) or $(q - 1)(r + 1) \le nd_B - 1$ at $r \ge 2$ (q a prime power). Note that the median obstruction requires five* coordinates*, not five outcomes: a single slot with $d_B \ge 5$ basis flags already triggers it, which is why the one-outcome state space and few-outcome quantum bodies are covered. On the constructive side: the complete classical dichotomy $\delta_r = 1 \iff n \le 2r + 2$ (Corollary 6.9); the complete qubit dichotomy $\delta_r = 1 \iff r \ge n - 1$ (Corollary 6.11, improving the two-sector pinching threshold $nQ_2(2) - 1 = 2n - 1$ and the earlier trace-mass threshold $r = n$); the exact boundary values $4/3$ (Corollary 6.7); and the exact value $2(1 - 1/d_B)$ at $r = n - 1$ for prime-power d_B (Theorem 6.12). At the bottom index the classification is complete (Corollary 6.16): $\delta_1 = 1$ exactly on $\{(n, d_B) : n \le 4,\ d_B =$*

$1\} \cup \{(1,2),(2,2)\}$ *(the latter by the mass-split encoder at $n = 2$) and $\delta_1 \geq 4/3$ on every other input-independent body; the two one-outcome cases left bracketed in the previous version are closed in the negative direction — the qutrit state space has $\delta_1 = 4/3$ exactly and the ququart lies in $[4/3,\, 3/2]$ (Theorem 6.14). For $d_B \geq 3$ above the diagonal at $r \geq 2$, flatness is certified from $r \geq nQ_2(d_B) - 1$ (Theorem 5.11) and the mass-split encoder gives $\delta_{n-1} \leq 2(1 - 1/d_B)$; the open range is $r \geq 2$ with $(nd_B - 3)/2 < r < nQ_2(d_B) - 1$, and by the concentration gate (Lemma 6.17) every flat instance at output dimension d_B requires $\delta_r\big(\mathcal{D}(B)\big) \leq 1$, so the gating unknowns are the state-space widths themselves, the smallest being δ_2 of the qutrit. For $d_B = 2$ below the diagonal, the mass-split encoder with $k \geq 1$ undecoded tail slots gives $\delta_{n-k} \leq 2 - 1/k$ (decode the tail blocks as $\big((1-s)/k\big)I_B/2$, where s is the decoded head mass; each tail block then errs by at most its trace plus its mass deviation from $(1-s)/k$), so the sub-boundary qubit widths lie in $[4/3,\, 2 - 1/k]$. Conjecture 6.19, in its corrected form, states the expected complete picture.*

Conjecture 6.19 (Coordinate dichotomy for input-independent flat widths)**.** *Assume $d_A = 1$ and $nd_B > 2$. Then*

$$\delta^{\diamond}_{r,\mathrm{inst}}(\mathbb{C}, B) = 1 \qquad \Longleftrightarrow \qquad nd_B \;\leq\; 2r+2 \quad \textit{and} \quad \big(d_B \leq 2 \textit{ or } r \geq 2\big) \quad \big(1 \leq r \leq nd_B^2 - 2\big).$$

The obstruction direction is proved for every $d_B \geq 1$: $nd_B \geq 2r+3$ forces $\delta_r \geq 4/3$ (Corollary 6.4, sharpened to $2(q-1)/q$ by Theorems 6.1 and 6.2 where a large enough q is available). The flat direction is proved for $d_B \in \{1,2\}$ (Corollaries 6.9 and 6.11) and fails at $r = 1$ for every $d_B \geq 3$ (Theorems 6.13 and 6.14): the earlier form of this conjecture, without the clause $r \geq 2$, predicted $\delta_1 = 1$ on the one-outcome bodies with $nd_B \leq 4$, i.e. exactly the two residual state-space cases, and is refuted — the qutrit has $\delta_1 = 4/3$ exactly and the ququart at least $4/3$ — which is why the clause $r \geq 2$ appears in the corrected statement. The content of the conjecture is therefore the flat direction at $d_B \geq 3$ with $r \geq 2$: that $r \geq \lceil (nd_B - 2)/2 \rceil$ latent dimensions — one per two flag coordinates — suffice for error one. This is a strong demand: the pinching constructions certify flatness only from $nQ_2(d_B) - 1$ upward, a rate of $Q_2(d_B)$ latent dimensions per slot. But it is the exact generalisation of both proved dichotomies, and it is consistent with the exact values of Theorem 6.12: for $d_B \geq 3$, the index $r = n-1$ lies strictly below the conjectured flat range (indeed $nd_B > 2n$) and there $\delta_{n-1} = 2(1 - 1/d_B) > 1$.

Example 6.20 ($d_A = d_B = 2$, $n = 2$)**.** *Here $D = 28$, $L_{\mathrm{rep}} = 6$, $L_{\mathrm{obs}} = 2$, $N_{\mathrm{join}} = 2$, $L_{\mathrm{join}} = 3$, so $L^{(n)}_{A,B} = 6$; $\rho_n = 1/4$ and $\gamma^{(n)}_{A,B} = \max\{1/4, 1/2\} = 1/2$. Thus*

$$\beta_r^{\mathrm{best},(n)} = \begin{cases} 1, & 0 \leq r \leq 6, \\ 1/2, & 7 \leq r < 28, \\ 0, & r \geq 28. \end{cases}$$

The upper envelope: $R_n = 1.75$; the convex bound equals $56/29 \approx 1.931$ at $r = 0$ and $34/18 \approx 1.889$ at $r = 11$, both above 1.75; balanced two-sector output pinching gives 1 at $r_{\mathrm{out}} = d_A^2(nQ_2(d_B) - 1)$, which evaluates to $4(2 \cdot 2 - 1) = 12$ (here $Q_2(2) = 2$); input pinching enters only at $r_{\mathrm{in}} = Q_2(d_A)(nd_B^2 - 1)$, which evaluates to $2(8-1) = 14$ and certifies the same value 1 from $r = 14$ on (Figure 6(a)). Hence on $0 \leq r \leq 14$ the verified upper envelope is 1.75 for $0 \leq r < 12$ and 1 for $12 \leq r \leq 14$. At $r = 0$ the exact width is $1.75 = \beta_0 = U_0$: the upgraded envelope value $\beta_0 = R_n$ meets the upper envelope there, while the one-sphere certificate alone would leave the residual gap $0.75 = 1 - \rho_n$. The envelope is plotted in Figure 6(a), and Table 3 collects the threshold data for several parameter triples (its $(3,2,2)$ row corresponds to Figure 6(b)).

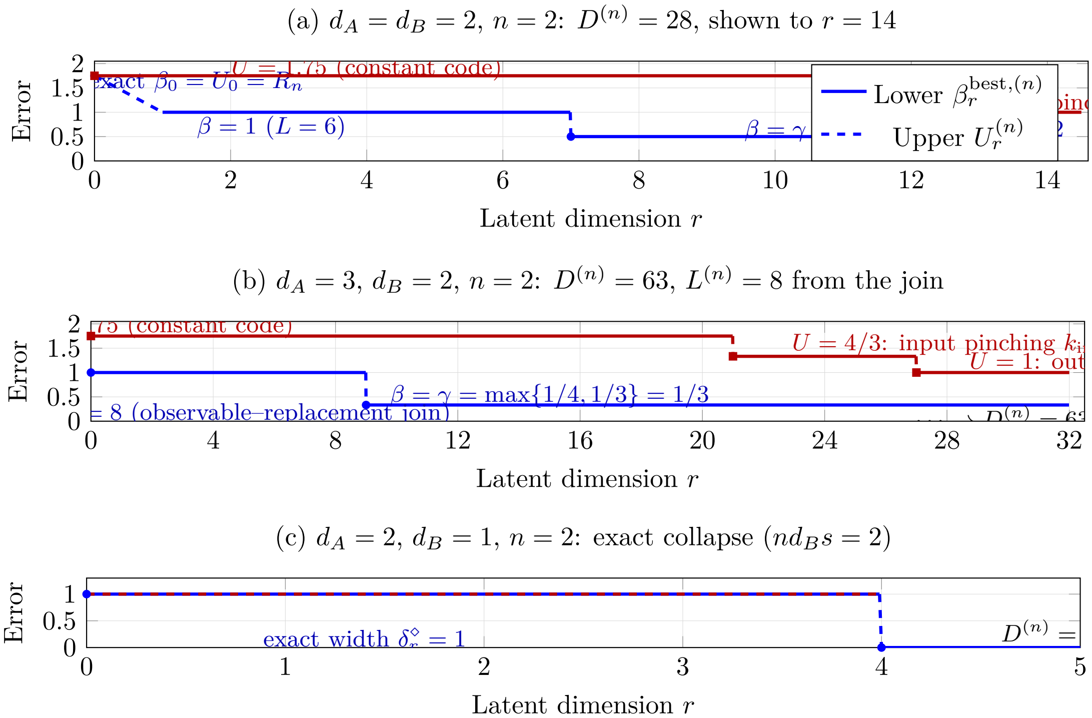

Figure 2: Certified lower and upper envelopes of the instrument compression width. (a) The worked example $d_A = d_B = 2$, $n = 2$ (Example 6.20): the upgraded value $\beta_0 = R_n = 1.75$ at $r = 0$ (exact, meeting the upper envelope) and $\beta_r^{\text{best},(n)} = 1$ for $1 \le r \le 6$, with $\gamma = \max\{\rho_n, 1/d_A\} = 1/2$ for $7 \le r < 28$; the upper envelope is $R_{\text{inst}} = 1.75$ for $0 \le r < 12$ and 1 from balanced two-sector output pinching for $12 \le r \le 14$, while convex projection exceeds 1.75 throughout and input pinching enters at $r_{\text{in}} = 14$ with the same value 1. (b) $d_A = 3$, $d_B = 2$, $n = 2$: the one-sphere dimension $L^{(n)} = 8$ comes from the observable–replacement join; the upper envelope steps down $1.75 \to 4/3 \to 1$ at $r = 21$ (three-block input pinching) and $r = 27$ (two-block output pinching), and the intermediate lower bound is $\gamma = 1/3$. (c) The exact collapse for $d_A = 2$, $d_B = 1$, $n = 2$ (Theorem 5.18): both envelopes equal the exact width, which is 1 below $D^{(n)} = 4$ and 0 from $D^{(n)}$ on. Filled circles and squares mark the lower and upper envelopes in (a) and (b); dashed segments mark the jumps.

7 Conclusion

The exact results of this paper are the affine dimension, the centred in-radius and its global optimality, the covering radius and the diameter, the constant-code width, the zero-error threshold, the replacement profile, and the collapse case, collected in Table 1. The structural fact behind this exactness is the concave–convex duality of the uniform instrument (Theorem 3.12 and Remark 3.13): the same object is optimal both as the centre of the largest contained ball and as the centre of the smallest covering ball. The intermediate widths, by contrast, are certified rather than exact: the lower envelope combines the two mechanisms of Remark 4.19, the upper envelope is attained by explicit constructions, and the bracket between them remains open in general.

The results specialise consistently. For $n = 1$ they recover the channel-level theorems of Ref. [2]; for $d_B = 1$ they govern finite-outcome measurements; and for $d_A = 1$ the antipodal profile of the full body is exact, so the top subcritical width and the one-sphere values of the envelope are determined exactly, while the input-independent subcritical widths obey the coordinate dichotomy for $d_B \in \{1, 2\}$ — $\delta_r = 1$ exactly above the diagonal $nd_B \le 2r + 2$ (classical: $n \le 2r + 2$; qubit: $r \ge n - 1$), $\delta_r \ge 4/3$ below it, with the exact boundary value

$4/3$ (Corollaries 6.9, 6.11, 6.7) — and the exact value $2(1-1/d_B)$ at $r=n-1$ for prime-power output dimensions (Theorem 6.12); at the bottom index the classification is complete — $\delta_1=1$ exactly on the classical $n\le 4$ and qubit $n\le 2$ bodies, $\delta_1\ge 4/3$ on every other, with the qutrit state space exactly at $4/3$ and the ququart in $[4/3,3/2]$ (Theorems 6.13 and 6.14, Corollary 6.16) — and the corrected coordinate Conjecture 6.19 covering the $d_B\ge 3$ flat direction at $r\ge 2$, gated by the state-space widths through Lemma 6.17. The $1/n$ suppression of the centred radius against the n-fold growth of the affine dimension (Corollary 5.21) quantifies the cost of recording outcomes, and the threshold of Theorem 5.18 identifies the cases in which the envelopes close.

The following problems remain open: (i) determine the intermediate widths $\delta^{\diamond}_{r,\mathrm{inst}}(A,B)$ for $L^{(n)}_{A,B}<r<D^{(n)}_{A,B}$, or improve either envelope; (ii) determine the largest index r with $\alpha_r(\mathrm{Inst}_n(A,B))=1$; (iii) resolve the boundary cases $c=d_B^2$ and $c=d_A^2d_B^2$ of the scaling trichotomy of Corollary 5.21; (iv) determine whether the uniform instrument is the unique optimal centre for the in-radius and for the covering radius; (v) settle the corrected coordinate flat-widths Conjecture 6.19 for $d_A=1$, $d_B\ge 3$ at $r\ge 2$ — by the concentration gate of Lemma 6.17 the gating unknowns are the state-space widths $\delta_r(\mathcal{D}(B))$, the smallest being δ_2 of the qutrit; determine the exact sub-diagonal widths, bracketed in $[2(q-1)/q,\ 2(n-2r-1)/(n-2r+2)]$ for $d_B=1$ (Theorems 6.1 and 6.2, Propositions 6.6 and 6.5(i)) and in $[4/3,\ 2-1/k]$ for $d_B=2$ at $r=n-k$ (Remark 6.18); and close the one-outcome ququart, bracketed in $[4/3,\ 3/2]$ (Theorem 6.14);

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A Finite-program evaluation: background

This appendix records, for completeness and motivation, the no-programming obstruction for unitary channels. It is logically independent of the instrument geometry developed above: the main theorems concern *representational* dimension, whereas this appendix concerns *programmable evaluation* with a fixed processor.

Definition A.1 (Exact finite-program evaluator)**.** *Let E and B be finite-dimensional. An exact finite-program evaluator consists of a finite-dimensional program system M, a completely positive trace-preserving (CPTP) processor $\Gamma:\mathrm{L}(M\otimes E)\to\mathrm{L}(B)$, and a program state $\pi_\Phi\in\mathcal{S}(M)$ for every channel $\Phi\in\mathrm{Chan}(E,B)$, such that*

$$\Gamma(\pi_\Phi\otimes\rho)=\Phi(\rho)$$

for every input state ρ on E. By linearity the same identity holds for every input operator.

Theorem A.2 (No finite exact evaluator for unitary channels)**.** *Let $\dim E>1$. No finite-dimensional program system and fixed CPTP processor $\Gamma:\mathrm{L}(M\otimes E)\to\mathrm{L}(E)$ can exactly evaluate every unitary channel on E.*

Proof. For pure program states, the no-programming theorem [13] shows that distinct unitary channels require orthogonal programs. It remains to pass from pure to mixed programs. Suppose

$$\pi_U=\sum_j q_j|p_j\rangle\langle p_j|,\qquad q_j>0,$$

is a spectral decomposition of a program that evaluates the unitary channel $\mathcal{U}_U$. For every pure input $|\psi\rangle$, linearity gives

$$U|\psi\rangle\langle\psi|U^\dagger=\sum_j q_j\,\Gamma\big(|p_j\rangle\langle p_j|\otimes|\psi\rangle\langle\psi|\big).$$

Because Γ is CPTP, each summand $\sigma_{j,\psi}$ is a density operator. The right-hand side is a convex decomposition of a pure state, and pure states are extreme points of the state space, so each positive-weight summand equals the same pure output: $\sigma_{j,\psi} = U|\psi\rangle\langle\psi|U^\dagger$ for all j and all pure $|\psi\rangle$. Fixing j, the map $\Lambda_j(X) = \Gamma(|p_j\rangle\langle p_j| \otimes X)$ is a channel agreeing with $\mathcal{U}_U$ on every rank-one projector; rank-one projectors span $\mathrm{Herm}(E)$ over $\mathbb{R}$, and complex linearity extends the equality to all of $\mathrm{L}(E)$. Hence every unit eigenvector of π_U with positive weight programs $\mathcal{U}_U$ exactly, and these eigenvectors span $\operatorname{supp}\pi_U$.

Applying the same argument to programs of two distinct unitary channels, the pure-state no-programming theorem makes every positive-weight eigenvector of the first program orthogonal to every positive-weight eigenvector of the second. Those eigenvectors span their respective supports, so $\operatorname{supp}\pi_U \perp \operatorname{supp}\pi_V$. A finite-dimensional program space cannot contain infinitely many mutually orthogonal nonzero supports, while $\dim E > 1$ gives infinitely many distinct unitary channels. This proves the claim. □

Corollary A.3 (Unequal output systems)**.** *Let* $\dim E > 1$ *and* $\dim B \geq \dim E$*. There is no finite-dimensional exact evaluator for all channels* $E \to B$*.*

Proof. Choose an isometry $V : E \to B$, let $\iota(\rho) = V\rho V^\dagger$, fix a state ω on E, and define the recovery channel

$$\mathcal{R}(\sigma) = V^\dagger \sigma V + \mathrm{Tr}[(I_B - VV^\dagger)\sigma]\,\omega,$$

which is CPTP and satisfies $\mathcal{R} \circ \iota = \mathrm{id}_E$. An evaluator for all channels $E \to B$ would in particular evaluate every channel $\iota \circ \mathcal{U}_U$ for the fixed embedding ι; postcomposing its processor with $\mathcal{R}$ would then give an exact finite evaluator for all unitary channels on E, contradicting Theorem A.2. □

Corollary A.4 (Vanishing error requires unbounded program dimension)**.** *Fix a system* E *with* $\dim E > 1$*. For each* j*, let* M_j *be a program system of dimension* m_j *and let* $\Gamma_j : \mathrm{L}(M_j \otimes E) \to \mathrm{L}(E)$ *be a CPTP processor. Write* $\Gamma_{j,\pi}(X) = \Gamma_j(\pi \otimes X)$ *and suppose*

$$\sup_{U \in \mathrm{U}(E)} \min_{\pi \in \mathcal{S}(M_j)} \|\Gamma_{j,\pi} - \mathcal{U}_U\|_\diamond \leq \varepsilon_j, \qquad \varepsilon_j \longrightarrow 0.$$

Then $m_j \to \infty$*.*

Proof. Suppose otherwise. Passing to a subsequence, assume $m_j \leq m_{\max}$ for every j. Let $M \simeq \mathbb{C}^{m_{\max}}$ and choose isometries $W_j : M_j \to M$. Put $P_j = W_j W_j^\dagger$, fix $\tau_j \in \mathcal{S}(M_j)$, and define

$$\Lambda_j(\eta) = W_j^\dagger P_j \eta P_j W_j + \mathrm{Tr}[(I_M - P_j)\eta]\,\tau_j,$$

a CPTP map $\mathrm{L}(M) \to \mathrm{L}(M_j)$: the first term is the compression onto the P_j-corner followed by the isometry $W_j^\dagger$, and the second is a measure-and-prepare channel on the orthocomplement (the two trace contributions add up to $\mathrm{Tr}\,\eta$). In particular $\Lambda_j(W_j \pi W_j^\dagger) = \pi$ for $\pi \in \mathcal{S}(M_j)$. The extended processors $\widetilde{\Gamma}_j = \Gamma_j \circ (\Lambda_j \otimes \mathrm{id}_E) : \mathrm{L}(M \otimes E) \to \mathrm{L}(E)$ are CPTP and preserve every originally programmed channel. The set of CPTP maps on the fixed spaces $M \otimes E \to E$ is compact, so, passing to a further subsequence *chosen independently of the target unitary*, $\widetilde{\Gamma}_j \to \Gamma$ in diamond norm. Now fix a unitary U. For each j choose $\pi'_j(U) \in \mathcal{S}(M_j)$ with approximation error at most ε_j (such a minimiser exists by compactness of $\mathcal{S}(M_j)$ and continuity of the induced channel), and embed it as $\pi_j(U) = W_j \pi'_j(U) W_j^\dagger \in \mathcal{S}(M)$. By compactness of $\mathcal{S}(M)$ there is a U-dependent subsubsequence with $\pi_j(U) \to \pi_U$; along it the processors still converge to the same Γ. Continuity of the evaluation map in both processor and program state gives $\Gamma(\pi_U \otimes \cdot) = \mathcal{U}_U$. Since U was arbitrary, the single limiting processor Γ exactly evaluates every unitary channel, contradicting Theorem A.2. Hence no bounded subsequence of (m_j) exists, and, (m_j) being integer-valued, $m_j \to \infty$. □

Remark A.5 (Scope of the no-programming obstruction). *The result forbids a scheme that is simultaneously exact, deterministic, universal over the unitary promise class, and finite-dimensional. It does not forbid approximate, probabilistic, restricted-family, infinite-dimensional, or unbounded-register schemes, nor schemes with an externally supplied classical description. A finite classical program register is still a finite-dimensional program and is not exempt. Storing a Choi operator as a mathematical state is also distinct from evaluating the represented channel with one fixed CPTP processor. The optimal asymptotic program cost of approximate universal unitary programming is determined in [14], and quantitative program-dimension lower bounds in the approximate case are obtained in [4]; the compactness corollary above records the qualitative divergence consequence needed for motivation. Complementary quantitative statements — the positive finite-programming floor, program-state separation, and the packing lower bounds — are developed in Ref. [2].*

Data Availability

No data were generated or analysed in this study.

References

[1] M.-D. Choi, *Completely positive linear maps on complex matrices*, Linear Algebra Appl. **10**, 285–290 (1975).

[2] [Authors anonymized for review], *Exact diamond balls, antipodal widths, and query-uniform compression bounds for quantum channels*, submitted for publication (2026). (Submitted concurrently with the present work.)

[3] A. Jamiołkowski, *Linear transformations which preserve trace and positive semidefiniteness of operators*, Rep. Math. Phys. **3**, 275–278 (1972).

[4] M. Kubicki, C. Palazuelos, and D. Pérez-García, *Resource quantification for the no-programing theorem*, Phys. Rev. Lett. **122**, 080505 (2019).

[5] E. B. Davies and J. T. Lewis, *An operational approach to quantum probability*, Commun. Math. Phys. **17**, 239–260 (1970).

[6] D. Aharonov, A. Kitaev, and N. Nisan, *Quantum circuits with mixed states*, in Proceedings of the Thirtieth Annual ACM Symposium on Theory of Computing (STOC 1998), ACM, New York, 1998, pp. 20–30.

[7] M. Ozawa, *Quantum measuring processes of continuous observables*, J. Math. Phys. **25**, 79–87 (1984).

[8] A. Pinkus, *n-Widths in Approximation Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete **7**, Springer, Berlin, 1985.

[9] G. Chiribella, G. M. D'Ariano, and P. Perinotti, *Realization schemes for quantum instruments*, J. Math. Phys. **50**, 042101 (2009).

[10] G. Chiribella, G. M. D'Ariano, and P. Perinotti, *Quantum circuit architecture*, Phys. Rev. Lett. **101**, 060401 (2008).

[11] C. W. Helstrom, *Quantum Detection and Estimation Theory*, Mathematics in Science and Engineering **123**, Academic Press, New York, 1976.

[12] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018).

[13] M. A. Nielsen and I. L. Chuang, *Programmable quantum gate arrays*, Phys. Rev. Lett. **79**, 321–324 (1997).

[14] Y. Yang, R. Renner, and G. Chiribella, *Optimal universal programming of unitary gates*, Phys. Rev. Lett. **125**, 210501 (2020).

[15] J. Matoušek, *Using the Borsuk–Ulam Theorem*, Springer (2003).

[16] A. Barvinok, *A Course in Convexity*, Graduate Studies in Mathematics **54**, AMS (2002).

[17] M. Özaydin, *Equivariant maps for the symmetric group*, unpublished manuscript, University of Wisconsin–Madison (1987).

[18] H. Tverberg, *A generalization of Radon's theorem*, J. London Math. Soc. **41**, 123–128 (1966).

[19] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics **87**, Springer, New York, 1982.

[20] J. W. Milnor and J. D. Stasheff, *Characteristic Classes*, Annals of Mathematics Studies **76**, Princeton University Press, Princeton, 1974.

[21] W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts **35**, Cambridge University Press, Cambridge, 1997.

Regime	Index range	Certified lower $\beta_r^{\mathrm{best},(n)}$	Certified upper $U_r^{(n)}$	Status
Constant code	$r=0$	$R_n = 2\big(1 - \frac{1}{nd_B s}\big)$	R_n	exact width $\delta^{\diamond}_{0,\mathrm{inst}} = R_n$ (Corollary 5.16); the envelopes coincide at $r = 0$, with one-sphere residual $R_n - 1 = 1 - \rho_n$
One-sphere regime	$0 < r \le L_{A,B}^{(n)}$	1	$U_r^{(n)} \ge 1$	exact at the top subcritical index $r = nd_B^2 - 2$ for the replacement subfamily, and for the full body when $d_A = 1$ (Corollary 5.17); for $d_A = 1$: the coordinate dichotomy — $\delta_r \ge 4/3$ when $nd_B \ge 2r+3$ (Corollary 6.4), $\delta_r = 1$ when $d_B = 1$, $n \le 2r+2$ (Corollary 6.9) and when $d_B = 2$, $r \ge n-1$ (Corollary 6.11); exact values 4/3 on the boundary $nd_B = 2r+3$ (Corollary 6.7) and $2(1 - 1/d_B)$ at $r = n-1$ for prime-power d_B (Theorem 6.12); at $r = 1$: $\delta_1 = 1$ iff $nd_B \le 4$ and $d_B \le 2$, with $\delta_1 \ge 4/3$ on every other input-independent body and $\delta_1 = 4/3$ on every state space of dimension ≥ 3 (Theorems 6.13 and 6.14, Corollary 6.16)
Intermediate	$L_{A,B}^{(n)} < r < D_{A,B}^{(n)}$	$\gamma_{A,B}^{(n)} = \max\{\rho_n, \frac{1}{d_A}\}$	$U_r^{(n)} \ge 1$	certified bracket of length at least $1-\gamma_{A,B}^{(n)}$; the true width is not determined in general (Proposition 5.19)
Full dimension	$r \ge D_{A,B}^{(n)}$	0	0	exact: $\delta^{\diamond}_{r,\mathrm{inst}} = 0$ (Theorem 5.7)
Collapse $nd_B s = 2$	$0 \le r < D_{A,B}^{(n)}$	1	1	exact: $\delta^{\diamond}_{r,\mathrm{inst}} = 1$ below $D_{A,B}^{(n)}$ and 0 from $D_{A,B}^{(n)}$ on (Theorem 5.18)

Table 1: Summary of the compression widths of the instrument body ($nd_B > 1$). The lower and upper envelopes coincide at $r = 0$ (both equal the exact constant-code width R_n of Corollary 5.16; Definition 4.17), in the collapse case, and from $r = D_{A,B}^{(n)}$ on; in the intermediate regime only the certified bracket is determined.

Quantity	Channel $n=1$	Instrument $n \geq 1$	Origin
Affine dimension	$d_A^2(d_B^2-1)$	$d_A^2(nd_B^2-1)$	$nd_A^2 d_B^2 - d_A^2$ via surjectivity of the marginal map $(X_1,\dots,X_n) \mapsto \sum_k \mathrm{Tr}_B X_k = H$, e.g. $X_1 = H \otimes I_B/d_B$
Centred in-radius	$2/d_B s$	$2/nd_B s$	$\max\{0, \lambda_{\max}(-X)\} \leq s\|\vec{\Delta}\|_{\diamond,\mathrm{inst}}/2d_A$, $s = \min\{d_A, d_B\}$
Covering radius $R(\mathcal{I}_*)$	$2(1-1/d_B s)$	$2(1-1/nd_B s)$	$\sigma_{EBC_n} \leq nd_B s\,(\rho_E \otimes I_B/d_B \otimes I_{C_n}/n)$ plus $a \leq Kb$
Replacement sphere	$S^{d_B^2-2}$	$S^{nd_B^2-2}$	$X \mapsto X_+/t_X$, orthogonal Jordan supports
Full-direction sphere	$S^{d_A^2(d_B^2-1)-1}$	$S^{D^{(n)}-1}$	Jordan decomposition plus correction state $\nu = (I_{A_0}-\mu)/(d_A-1)$ for $d_A \geq 2$; for $d_A = 1$ set $J^{\pm} = \Theta_{\pm}$
Diamond diameter	2	2	flagged channels have diamond norm 1; orthogonal flagged states attain 2
Zero-error threshold	$d_A^2(d_B^2-1)$	$d_A^2(nd_B^2-1)$	affine coordinates plus Borsuk–Ulam collision

Table 2: Channel versus instrument exact geometry. The $1/n$ factor is the uniform trace-budget split among n outcomes at the uniform centre; the same numerical positivity budget $2/(nd_B s)$ is realised by two distinct witnesses — an orthogonal-output isometry witness for $d_B > 1$, and an effect-transfer witness $(\pm\mathcal{F}/2)$ for $d_B = 1$, $n \geq 2$ — which live in different faces of the positive cone. Maximality of the radius uses the eigenvalue $1/(nd_A d_B) - s\,t'/(2d_A) < 0$ for $t' > t$, or $1/(nd_A) - t'/(2d_A) < 0$ in the POVM case. The sphere rows are stated in the non-singleton regimes ($d_B > 1$ for channels, $nd_B > 1$ for instruments); the $n = d_B = 1$ singleton is described in Remark 2.5.

(d_A, d_B, n)	$D^{(n)}$	L_{rep}	L_{obs}	N_{join}	L_{join}	$L_{A,B}^{(n)}$	$\gamma_{A,B}^{(n)}$
$(2,2,2)$	28	6	2	2	3	6	1/2
$(3,2,2)$	63	6	7	2	8	8	1/3
$(2,3,1)$	32	7	2	1	2 (degenerate)	7	1/2
$(2,1,2)$	4	0	2	0	invalid	2	1

Table 3: Certified one-sphere dimensions and intermediate envelope value, with $L_{A,B}^{(n)} = \min\{D^{(n)}-1, \max\{L_{\mathrm{rep}}, L_{\mathrm{obs}}, L_{\mathrm{join}}\}\}$ and $\gamma_{A,B}^{(n)} = \max\{\rho_n, 1/d_A\}$. For $(2,1,2)$ one has $nd_B s = 2$ and the collapse of Theorem 5.18 applies: $\beta_r^{\mathrm{best}} = 1$ throughout $r < D$.